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CEO Social Capital, Risk-Taking and Corporate Policies

by

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Abstract

We provide the first direct empirical evidence of the effect of CEO social capital on aggregate corporate risk-taking. Our theory predicts that CEOs with high social capital display higher levels of risk-seeking behavior. Consistent with this prediction, we find a positive association between CEO social capital and aggregate corporate risk-taking. Examining the channel, we show that social ties cause corporate policy actions, and these actions lead to greater volatilities in stock returns and earnings. In addition, we uncover a number of factors that significantly moderate the effects of social capital on risk-taking. We also show that this increase in risk-taking is value-enhancing to the firm. Our results are robust to alternative proxies for risk-taking, alternative model specifications, and tests for endogeneity.

JEL classification: G30, Z13

Keywords: Social capital, social networks, corporate risk-taking

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1. Introduction

This paper examines the effects of CEO social capital on corporate risk-taking. More specifically, we ask whether CEO social capital influences risk-taking incentives, and how it is manifested in corporate policies. As a related question, we examine the implications of corporate excess risk-taking that is directly attributable to CEO social capital for firm valuation.

The importance of these questions stems from the contemporaneous corporate finance research that shifts the emphasis on the individual attributes of CEOs, such as personal characteristics, attitudes, and preferences, and the influence these attributes exert on corporate policies. Research shows that CEO religion (Hilary and Hui, 2009), political preferences (Hutton et al., 2014), optimism and risk-aversion (Graham et al., 2013), sensation seeking (Cain and Mckeon, 2016) influence corporate behavior and risk-taking. We continue this line of inquiry by examining how a previously unexamined factor, the CEO's social capital, influences the riskiness of a firm's corporate policy choices.

Over the last decade, the concept of social capital has emerged as one of the most prominent topics in the modern social sciences (e.g., Putnam, 1993; Knack and Keefer, 1995; Woolcock, 1998; Portes, 1998; Fafchamps, 2002; Dasgupta, 2005; Schneider, 2006). We follow Woolcock (1998) and define social capital as information, trust, and norms of reciprocity inherent in a social network. Powerful agents such as the World Bank and various national governments pay significant attention to the idea of social capital as a strategy to mitigate market shortcomings. Yet the research on social capital has received only limited attention in academic finance. This paper reinforces a growing awareness among finance researchers that managerial social capital matters in corporate decisions.

Following the predictions of the theories of social capital and prior empirical evidence we argue that CEO social capital has implications for corporate risk-taking. We propose that social capital has a positive effect on corporate risk-taking for several reasons. First, social capital alters risk-tolerance of socially connected individuals because it offers a way to pool individual risks. For instance, Genicot and Ray (2005) and Bloch et al. (2008) show that social capital offers a mechanism of informal insurance which can increase risk-taking[1].

Second, social capital reinforces an individual's sense of power and as a result leads to riskier preferences. Rowley (1997) contends that with increased ties and access an individual's power within a social network is strengthened. The approach-inhibition theory of Keltner et al. (2003) argues that the experience of power drives people to take more risks.

Third, agency theory (e.g., Amihud and Lev, 1981; Hirshleifer and Thakor, 1992) posits that career concerns of executives are one of the major reasons for low managerial risk-taking. Social capital reduces information asymmetry in the labor market, and that way minimizes the consequences of failure to the CEO by providing labor market insurance against job losses. For example, Nguyen (2012) finds that socially connected CEOs are more likely to find new employment even after forced departures. Cingano and Rosolia (2012) show that the duration of unemployment declines with the quality of a worker's personal network.

In addition, we postulate that several magnifying (mitigating) factors could have implications for the effects of social capital on risk-taking. If social capital eases pressure of managerial labor markets by reducing the expected cost of dismissal, which in turn allows managers to pursue

[1]Based on Arnott and Stiglitz's (1991) informal insurance model, Mobarak and Rosenzweig (2013) show that informal insurance through social network structure leads more agricultural risk-taking in rural India. Miller and Paulson (2007) find that household gambling increases with the quality of informal insurance.

riskier corporate policies, then the effects of social capital should be more pronounced for firms with low entrenchment. We further posit that free cash flow, information asymmetry, external financing, as well as investment efficiency could significantly moderate the extent of influence of CEO social capital on corporate risk-taking.

To test these predictions, we use the BoardEx database provided by Management Diagnostics Limited for measuring our social capital proxies. BoardEx provides biographical information on current and past employment, the education, and other social activities of top corporate executives beginning in 1999. We use this biographical information to construct our social capital measure which represents the aggregation of these three aspects of social interactions: the number of individuals with whom the CEO shares a common educational, employment, or social history in BoardEx. Next, we follow prior research (e.g., Faleye et al. 2014) and estimate 1 plus the natural log of this aggregate social connections number.

Another key variable for our analysis is the CEO Human Capital Index, which is also constructed using the BoardEx database. Specifically, it is estimated as the sum of the following indicator variables: an indicator variable that takes the value of 1 if the CEO has an academic degree from an "elite" college; an indicator variable that takes the value of 1 if the CEO has a Ph.D.; an indicator variable that takes the value of 1if the CEO has legal experience; an indicator variable that takes the value of 1if the CEO has finance experience; an indicator variable that takes the value of 1if the CEO has political experience and an indicator variable that takes the value of 1if the CEO has military experience.

Our empirical results are consistent with theoretical predictions. We find that CEO social capital positively influences aggregate corporate risk-taking, where risk-taking is variously measured as the annualized standard deviation of monthly stock returns and earnings. Next, we examine the channels through which social capital exerts influence on aggregate risk-taking and find that CEO social capital increases riskiness of specific corporate investment and financial policies (such as, R&D expenditures, corporate diversification, financial leverage, and asset liquidity), which in turn, make future stock returns and earnings more volatile.

We also document that the increase in risk-taking attributable to the CEO's social capital has implication for risk-adjusted valuation. Further, we show that the effect of social capital on risk-taking is stronger for firms with low managerial entrenchment, low free cash flows, high external equity financing, high information asymmetry, and firms that previously underinvest.

Our results are invariant to alternative model specifications and estimation windows. The identification challenge of the paper is that it is possible that firms that generally engage in riskier corporate policies could employ well-connected CEOs. To address the potential endogeneity, we estimate instrumental variable two-stage least squares (2SLS) regressions using two different sets of instruments and still find supportive evidence of our main conjectures.

Our study makes several contributions to the literature. First, our study complements and extends the literature examining the effects of various CEO personal attributes on corporate risk-taking (e.g., Hilary and Hui, 2009; Hutton et al., 2014; Graham et al., 2013; Cain and Mckeon, 2016). We investigate an important and previously unaccounted for attribute, CEO social capital, and find its association with risk-increasing decisions.

Second, this study contributes to the emerging literature on the effects of social networks on corporate decision-making. Shue (2013) and Fracassi (2016) are the closest studies to our work. These authors examine the implications of the "peer effect" aspect of social connectedness on managerial decision-making and consequently for corporate policies. "Peer effect" is a tendency of individuals (and thus firms) to behave more similarly by following group leaders and adopting a group norm (e.g., Banerjee 1992, Scharfstein and Stein 1990). Fracassi (2016) explicitly shows that socially connected managers make more similar decisions. That is, the more connections that two companies share via executives, the more similar are their corporate policies (e.g., capital investments, R&D expenses, cash reserves, and interest coverage ratios). Using a natural experiment involving randomly assigned peer groups, Shue (2013) shows that executives who graduate from the Harvard Business School and are assigned to the same core-class section, adopt similar firm policies (with the strongest evidence of peer similarities occurring in executive compensation and acquisition policy). In contrast, to these studies, we focus on other aspects of social networks that can alter managerial risk tolerance. Even though the proxies for social connections are similar, our CEO social capital measure captures primarily labor market insurance effects. By providing a mechanism of informal insurance and reinforcing a sense of power, social capital leads to the adoption of riskier preferences, which in turn, is manifested in the increased riskiness of discretionary corporate policies.

Third, this study also adds to the literature on the impact of social networks in finance. For example, Hwang and Kim (2009) find that social connections between the CEO and the board lead to higher CEO pay, lower pay for performance sensitivity, and lower probability of CEO turnover. Cai and Sevilir (2012) show that social connections improve information flow between a target and an acquirer. Engelberg et al. (2012) find that social connections between banks and borrowers reduce borrowing costs. El-Khatib et al. (2015) show that better connected CEOs engage in more acquisitions. We find that CEO social networks increase risk-taking and lead to higher firm value.

Finally, we contribute to the literature examining the factors, such as governance, ownership structures, investor protection, which affect the riskiness of corporate policies (e.g., John et al., 2008; Laeven and Levine, 2009; Faccio et al., 2016). We find that CEO social capital increases the riskiness of firm investment and financial policies, and these policies channel the effects of social capital on aggregate corporate risk-taking.

The paper is organized as follows. Section 2 reviews the literature and develops empirical hypotheses. Section 3 describes our empirical model and variable construction. Section 4 presents our empirical results. Section 5 contains our robustness tests. Section 6 discusses our corrections for possible endogeneity. Section 7 examines the effect of risk-taking on firm value. Section 8 concludes with a brief summary and interpretation of our findings.

2. Prior Literature and Hypothesis Development

2.1. Social Capital Theories and Corporate Finance

The concept of social capital offers powerful insights into various socio-economic phenomena. We follow Woolcock (1998) and define social capital as the information, trust, and norms of reciprocity inherent in a social network, where the social network is the real-world links between groups or individuals. The central premise of social capital is that social networks have value.

We aggregate social capital theories into two broad groups: cognitive and structural. From the cognitive perspective, social capital can be understood as shared norms, attitudes, and beliefs (e.g., Coleman, 1988; Putnam, 1993). Structural theories advanced by Lin (1999, 2001) are based on Bourdieu's (1984, 1986, 1989) theoretical framework and focus on the pattern and intensity of various network connections. From these theories of social capital, we identify four distinct mechanisms through which social capital exerts an influence on corporate finance. These mechanisms are trust, the flow of information, the ability to punish and reward, and the ability to alter preferences.

A number of prior studies posit that trust is an important dimension of social capital (e.g., Dasgupta, 1988; Fukuyama, 1995). Trust minimizes negative consequences of incomplete contracts (Grossman and Hart, 1986) by reducing the need for costly monitoring. We argue that through this link social capital affects capital budgeting and firm-level risk-taking.

From informational perspective, social capital improves economic efficiency and enhances coordination thereby reducing information asymmetry. Recent finance and economics literature contains many applications of this information-based idea of social capital, though researchers are reluctant to use the term social capital. From the informational perspective, a number of studies, such as Rauch and Casella (2001), Cohen et al. (2008), Kuhnen (2009), Hochberg et al. (2010), Engelberg et al. (2012) provide consistent evidence that social capital through social networks opens new channels for the circulation of information, reduces the cost of external financing, expands stock market participation and plays an important role for portfolio choice decisions.

Information sharing is also an important driver of peer influence. Peer interactions can affect managerial decision-making via word-of-mouth effects (e.g., Ellison and Fudenberg, 1995; DeMarzo et al. 2003). Prior research shows that executive peer effects have clear implications for discretionary corporate policies. Fracassi (2016) finds that managers are influenced by their social peers when making corporate policy decisions. Specifically, the presence of social ties increases the similarity in capital investments (as well as R&D, cash reserves, and the interest coverage ratio). Further, Shue (2013) argues that "peer interactions could affect managerial decision- making because information and beliefs travel through social networks" and documents similarities in investment, compensation, leverage, interest coverage, and cash policies of socially connected firms via educational links (using the random assignment of MBA students to sections at Harvard Business School for identification).

Guaranteeing that accurate information is transferred through social networks requires the existence of a disciplinary mechanism. Social capital plays a critical role in circulating information about the breaches of contracts. This allows socially connected groups to penalize and exclude cheaters through reputation loss or penalty (e.g., Kandori, 1992; McMillan and Woodruff, 2000).

Finally, identification with a group or network can alter the preferences and choices of connected individuals. Through this channel, social capital affects corporate risk-taking. Social capital may stimulate riskier preferences by pooling individual risks. This argument is based on the growing literature studying informal insurance within a social network (e.g., Genicot and Ray, 2005; Bloch, Bramoulle and Kranton, 2007; Genicot and Ray, 2008).

2.2. Hypothesis Development

We begin this subsection by posing the following question: Does the social capital of a CEO influence decisions about corporate risk-taking, and if so how is it manifested in corporate policies? There are three arguments that justify a positive relation between corporate risk-taking and the CEO's social capital.

First, social capital offers a way to pool individual risks and through this channel affect the risk-tolerance of socially connected individuals. Recent research shows that social networks are a mechanism for risk-sharing. Allen and Gale (1997), Acemoglu and Zilibotti (1997), and Ambrus et al. (2014) argue that better risk-sharing tends to promote more risk-taking, consequently leading to firm growth. A defining feature of the individual decision-making problem is dealing with risk. Miller and Paulson (2007) argue that uninsured risk can cause households to make inefficient choices and suggest that households allocate resources more efficiently when better insurances are available. Because social capital offers a mechanism of informal insurance (Genicot and Ray, 2005; Bloch et al., 2008), it can increase CEO risk-taking.

Second, social capital intensifies an individual's sense of power, leading to riskier managerial preferences and consequently increased corporate risk-taking. Managerial power can stem from executives' personal links to institutional investors, suppliers, and other economic agents. These social networks generate channels of information-sharing, resource acquisition, and alternative employment opportunities for the connected individuals. In addition, personal contacts with representatives of other organizations create valuable resources for managers, such as door-opening and legitimizing (Borch and Huse, 1993). Socially well-connected executives represent a "special segment of the capitalist class" with significant power due to the multiple connections to large corporations (Useem, 1979). An executive's power grows as the individual gains more ties and access (Rowley, 1997). Therefore, the CEO's social capital is a source of power and prestige for the individual.

We apply the approach-inhibition theory of power (Keltner et al., 2003) from social psychology to demonstrate that power can drive managers to take more risk. This theory argues that power causes those who possess it to focus on the potential reward aspects of risky behavior and thus to greater risk-taking. Testing this theory, Anderson and Galinsky (2006) find empirical evidence of a positive relation between power and risk-taking by individuals. The study implies that the effect of power on risk-taking is not associated with self-efficacy. The findings show that powerful people focus more on the potential payoffs associated with risk-taking while focusing less on the possible accompanying dangers. Additionally, power increases optimism in the presence of risk, which consequently leads to an increased propensity for risk-taking.

Third, agency-based theoretic models show that one of the major reasons for low managerial risk-taking is the career concerns of executives (Amihud and Lev, 1981; Holmstrom and Ricart Costa, 1986; Hirshleifer and Thakor, 1992). Social capital is an efficient instrument for achieving better employment outcomes. Research by Holzer (1987), Burt (1992), Granovetter (1995), and Calvo-Armengol and Jackson (2004) establishes that social networks can overcome the information asymmetry that exists in the labor market by transmitting crucial information, such as news about job vacancies or accounts of workers' abilities. A number of studies (e.g., Granovetter, 1973; Zhou, 1992; Schneider, 2006) argue that social capital is instrumental for access to jobs and is related to more successful employment outcomes. Nguyen (2012) explicitly shows that socially connected CEOs are more likely to find reemployment after a (forced) departure. Faleye et al., (2014) find that CEO personal connections offer implicit labor market insurance that mitigates the career concerns by enhancing the prospect for re-employment. This analysis leads to the prediction that because a potential loss of employment is a personal cost borne by the CEO and social capital alleviates these concerns it should have a positive effect on the corporate risk-taking.

Next, we argue that CEO social ties cause specific, more risky, corporate policy actions, and these actions lead to greater aggregate risk-taking. More specifically, we predict that well-connected CEOs are likely to prefer investment and financial policies that are more risky. We follow Cassell et al. (2012) and focus on two investment policy mechanism through which CEOs can increase aggregate risk-taking: First, research and development (R&D) expenditures are more risky (e.g., Bhagat and Welch, 1995) compared to other investment choices. Second, CEOs could increase riskiness of their firms by reducing diversification across different industry segments (e.g., Amihud and Lev, 1981; Coles et al., 2006).

We also posit that CEOs with high social capital may adopt riskier financial policies. More specifically, CEOs could amplify risk exposure by increasing debt burden of the firm (e.g., Cassell et al., 2012) or by holding less liquid assets. Consistent with this analysis, we predict a positive association between CEOs social capital and R&D expenditure, negative association between CEO social capital and corporate diversification. Further, we argue that CEO connectedness increases in financial leverage and decrease in asset liquidity. In addition, we posit that these policies channel the effects of CEO social capital on aggregate risk-taking.

Further, we propose that several important second-order magnifying (mitigating) factors could have implications for the connectedness-risk relation. First, if social capital indeed alleviates the pressure of managerial labor markets, and consequently allows managers to pursue riskier corporate policies (e.g., Faleyeet al., 2014), then this effect should be stronger for firms with low managerial entrenchment. On the other hand, for high entrenchment firms the threat of dismissal is much smaller, and thus connectedness-risk link should be weaker. Second, potential high spending by high free cash flow firms could be inefficient and value reducing. If social capital enhances economic efficacy and is associated with optimal increase in risk-taking, then its effects should be more pronounced for firms with low free cash flows. Third, firms with high information asymmetry could find it difficult to persuade investors to fund high-risk projects. Consequently, we propose that social capital, given that it alleviates information asymmetry, should have stronger implications for risk-taking for firms with greater information asymmetry. In addition, social capital effects could also be moderated by external financing[2]. Finally, we also postulate that the implications of social capital for risk-taking could be conditional on investment

[2] Javakhadze et al. (2016) explicitly show that social capital exerts influence on external finance sensitivity to Q.

efficiency. Specifically, if firms underinvest, the effects of social capital should be stronger, compared to firms that overinvest.

Managers, due to their career concerns and lack of diversification, are more risk-averse than shareholders, which is sub-optimal for the firm (Jensen and Meckling, 1976; Demsetz and Lehn, 1985). If social capital induces managers to take more risk and therefore better aligns managerial and shareholder interests, we should observe a positive relation between risk-taking attributable to social capital and firm performance. However, El-Khatib et al. (2015) actually show that highly connected CEOs initiate excessive risky investment (M&A) which is value-reducing. Consequently, higher risk due to higher connectedness may not unambiguously lead to higher value. Therefore, we propose that, conditional on managerial entrenchment, free cash flow, external financing, information asymmetry, and investment efficiency, increased risk-taking associated with social capital could be value-enhancing.

3. Sample Selection, Model Specification and Variable Construction

3.1. Data and Sample selection

Our empirical analysis requires data obtained from a variety of databases. We obtain data regarding social networks from the *BoardEx* database of Management Diagnostic Limited. BoardEx provides social network data for senior executives and directors for a set of global public and private firms since 1999[3]. This establishes the starting point for our empirical analysis as 1999. From the profiles, we collect information on current employment, past employment, education background, and affiliation to professional associations, not-for-profit associations, and clubs. We match *BoardEx* data with *Compustat* to obtain the corresponding financial and

[3] Social ties formed before 1999 are also captured because BoardEx records the covered executives and directors' career, education, and activity history for as long as data are available.

accounting variables. We obtain industry segment data from the Compustat segment data file. Firm-level return data are from the Center for Research in Security Prices (*CRSP*).

We obtain data necessary to calculate compensation wealth effects as well as overconfidence measure from *Execucomp*. Data for estimating managerial entrenchments measures is obtained from Institutional Shareholder Services (formerly RiskMetrics) database[4]. We obtain board co-option and hostile takeover threat measures from Lalitha Naveen's and Stephen McKeon's websites, respectively. County-level data on social organizations are from Northeast Regional Center for Rural Development, Penn State (Rupasingha et al., 2006). Data for systematic market risk, size, market-to-book ratio, and momentum factors as well as the risk-free rate are obtained from Kenneth French's website. In our main tests we use the volatility measures constructed over the window t+1 through t+3. Due to this additional requirement, our sample period ends in 2011. The final sample consists of 29,748 CEO-firm-year observations for the period 1999-2011.

3.2. Empirical Model and Measurement of Main Variables

To test our hypotheses, we follow prior studies (e.g., Faccio et al., 2016; Cassell et al., 2012; Cain and McKeon, 2016) and develop a model to examine the effects of CEO social capital on aggregate corporate risk-taking as well as on the riskiness of firm investment and financing policies:

Risk-taking(Corporate Policies)=f(CEO Social Capital, Controls) (1)

The major variable of interest is our measure of CEO social capital. Our main hypotheses predict that social capital positively affects aggregate corporate risk-taking choices and riskiness of corporate policies.

3.2.1 Measuring Aggregate Corporate Risk-Taking

[4] We obtain board co-option and hostile takeover threat measures from Lalitha Naveen's and Stephen McKeon's websites, respectively.

Motivated by prior work, we consider three measures of corporate risk-taking. Our primary measures of corporate risk-taking is *Volatility (Return)* - volatility of monthly stock returns[5] constructed over the window *t*+1 to *t*+3 (e.g., Cassel et al., 2012; Becker and Strömberg, 2012; Cain and McKeon, 2016). Firms with high return volatility are riskier than firms with low return volatility. We favor this measure because the accounting measures of risk-taking could be subjected to manipulations. In robustness tests we estimate return volatility over the window *t*+1 through *t*+5.

We also use two accounting performance volatility proxies for risk-taking commonly used in the literature: the volatility of the return on assets (*Volatility (ROA)),* where the return on asset is the ratio of operating income to total assets, and the volatility of a firm's return on equity (*Volatility (ROE)),* where the return on equity is the ratio of operating income to total stockholders' equity. The intuition behind these risk-taking measures is that riskier corporate operations lead to more volatile earnings (e.g., John et al., 2008; Faccio et al., 2016). In addition, the volatility of ROE reflects both the riskiness of a firm's projects and the additional risk induced by the use of leverage in capital structure. We estimate both the *Volatility (ROA)* and *Volatility (ROE)* by using quarterly data over three-year overlapping periods in our main tests, and over five-year overlapping periods in the robustness tests. These measures of risk-taking capture the overall risk taken by the firm.

3.2.2 Measuring CEO Social Capital

Empirical measures of the social capital used in this study are based on the structural theories of social capital of Bourdieu (1984, 1986), Burt (1992), and Lin (1999). In these theories social capital is sometimes referred to as social network capital to emphasize that individuals

[5] We also repeat the tests with daily returns and obtain qualitatively similar results. In addition, our results are qualitatively similar if idiosyncratic risk, estimated as the volatility of residual returns over market model (e.g., Xu and Malkiel, 2003; Cassell et at., 2012) is used as a measure of risk-taking.

derive benefits from knowing others with whom they form networks. In this view, social capital is an embedded resource that is accessed through network membership. This view of social capital makes it empirically measurable.

To measure CEO social capital, we first follow prior research of Freeman (1979) and Burt (1983) and estimate the centrality measure of CEO social networks. We count the number of individuals with whom the CEO shares common educational, employment, or social history in the BoardEx[6] database. Next, we follow Faleye et al. (2014) and estimate 1 plus natural log of this number, which is our main measure of CEO social capital (*SC(Total)*). Our CEO social capital measure is estimated at the beginning of the period (t) over which volatilities are estimated[7].

CEO social capital may also be affected by human capital. To account for the possibility that the link between corporate risk-taking and CEO social capital may be due to CEO's superior human capital, we filter the human capital out of our CEO social capital measure and use the excess connectedness - $SC(Total)_{excess}$ (estimated as the residuals from regression of CEO social capital on CEO human capital index) in our robustness tests as an alternative measure of CEO social capital. *CEO Human Capital Index* is estimated as the sum of the following indicator variables: an indicator variable that takes the value of 1 if the CEO has an academic degree from "elite" college as defined by Engelberg et al. (2013) and zero otherwise; an indicator variable that takes the value of 1 if the CEO has a Ph.D., and zero otherwise; an indicator variable that

[6] Following Faleye et al. (2014) we exclude connections derived from the CEO's employment as CEO at his current firm. We note that our results are qualitatively similar if we use detrended measure of CEO social connections.
[7] We also carve several components out of CEO social capital: SC(Educ) - the number of individuals with whom the CEO attended the same university, graduated within 2 years of each other, and earned a similar type of degree; SC(Empl) - the number of individuals with whom the CEO shares common employment history; and SC(Other) - the number of individuals with whom the CEO is connected through social clubs or other non-profit organizations. In addition, we also separately measure the CEO's social capital with financiers and S&P 500 firms. We define "financiers" as investment firms classified by BoardEx as "investment companies," "private equity," "specialty and other finance," "bank" or "insurance company.

takes the value of 1if the CEO has legal experience, and zero otherwise; an indicator variable that takes the value of 1if the CEO has finance experience, and zero otherwise; an indicator variable that takes the value of 1if the CEO has political experience, and zero otherwise; and an indicator variable that takes the value of 1if the CEO has military experience, and zero otherwise[8].

3.2.3 Measuring Riskiness of Corporate Policies

We measure riskiness of both the investment and financial policies. We use two proxies to capture the riskiness of investment policies: R&D expenditures and corporate diversification. We follow prior studies (e.g., Opler and Titman, 1994; Bebchuk and Cohen, 2005; Cassel et al., 2012) and define our R&D expenditures as the ratio of research and development expenses to total sales (*R&D/Sales*). We estimate the extent of corporate diversification by using the entropy measure (Jacquemin and Berry, 1979). We calculate *Diversification* as $Entrophy = \sum P_s \log(1/P_s)$, where P_s is the proportion of a firm's assets in the industry segment s[9].

We adopt two measures of the riskiness of firm financial policies (e.g., Cassell et al., 2012). First, we proxy for the debt burden in the capital structure by *Leverage* estimated as the ratio of total debt to total assets. Second, we capture the liquidity of the firm's assets by *Working Capital* estimated as current assets minus current liabilities scaled by total assets[10]. Parallel to our risk-taking proxies, we estimate our measures of the riskiness of corporate policies as the average over the window *t*+1 to *t*+3 in the main tests, and as the average over the window *t*+1 to *t*+5 in the robustness tests.

[8] The construction of this index follows Qualifications Index from Fedaseyeu et al. 2016.

[9] When segment data are not available, we assume the firm's assets are in one segment. We opt to estimate our diversification measure based on assets because it better proxies for risk in investment policies than sales. However, our results are qualitatively similar if sales are used instead.

[10] Following Cassell et al. (2012), when the values of these measures are missing we estimate working capital as the sum of cash and short-term investments, other current assets, inventory, and current accounts receivables, minus the sum of accounts payable, other current liabilities, debt in current liabilities, and income tax payable.

3.2.4 Measuring Other Variables

To control for correlated omitted variables, we augment our regression equations with control variables motived by prior research (e.g., John et al., 2008; Faccio et al., 2016; Cassell et al., 2012; Cain and McKeon, 2016). We include firm age (*Log(Age)* - the natural logarithm of the age of the firm) and size (*Log(Size)* - the natural log of total assets) to account for the systematic variations in firm riskiness related to the lifecycle (Pastor and Veronesi, 2003). We control for growth and investment opportunities by including market-to-book ratio and sales growth because high-growth firms are inclined to take more risk (Coles et al., 2006). We estimate market-to-book ratio *(MB Ratio)* as the ratio of the market value of equity to the book value of assets, and growth in sales (*Sales Growth)* as the ratio of total sales to lagged total sales.

We also include the stock return over the fiscal year t (*Return*) because past performance could have implications for firm riskiness. We control for the historical financing decisions and internal capital availability to finance investments by *Debt/Equity Ratio* estimated as the ratio of total debt to the market value of equity, and by *Cash Surplus*, estimated as net cash flow from operations minus depreciation expense plus research and development expenditures, scaled by total assets.

A number of studies such as Cohen et al. (2000), Coles et al. (2006), Chakraborty et al. (2007), Hirshleifer et al. (2012), and Cain and Mckeon (2016) include CEO tenure and CEO age to proxy for CEO risk-aversion. Managerial age is also used as a proxy for career concerns (Gibbons and Murphy, 1992). Consequently, we use *Log(CEOAge)* and *Log(CEOTenure)* as additional control variables.

We use managerial entrenchment, free cash flow, external financing, information asymmetry and investment efficiency measures as moderating variables. We proxy for managerial entrenchment by *CEO Pay Slice* (estimated as the fraction of the aggregate compensation of the top-five executive team captured by the CEO) as well as by *GIM Index* from Gompers et al. (2003), *BCF Index* from Bebchuk et al. (2009), *Co-option* from Coles et al. (2014), and *Hostile Takeover Threat* measure from Cain et al. (2017). *Free Cash Flow* is estimated following Lehn and Poulsen (1989). *External Financing* is the annual change in book value of equity plus the change in deferred taxes minus the change in retained earnings, all scaled by lagged assets. *Information Asymmetry* is measured as the absolute value of analysts' earnings forecast error for the fiscal year, estimated as the absolute value of actual earnings minus the earnings forecast scaled by the stock price. *Investment Efficiency* is estimated as the residual from a simple investment model (Biddle et al., 2009) that predicts the level of investment based on growth opportunities (measured by sales growth). Positive residual, as reflected in the error terms of the investment model, represents overinvestment and negative residual – underinvestment.

In our robustness tests we include additional controls, *Asset Tangibility* and *Free Cash Flow*, to account for the potential of financial distress and availability of funds to invest in new projects. As suggested by Guay (1999) and Cassel at el. (2012), we control for the CEO's outside wealth and the degree of diversification by *Log(Total Comp)* - the natural logarithm of the sum of salary and bonus compensation. Because the incentives arising from the equity-based compensation are important determinants of the risk-seeking behavior (Rajgopal and Shevlin, 2002; Coles et al., 2006), we include *Delta* - the sensitivity of the value of the CEO's accumulated equity-based compensation to a one-percent change in the stock price, and *Vega* - the sensitivity of the value of the CEO's accumulated equity-based compensation to a one-percent change in the volatility of stock prices as controls.

Further, there is an ample evidence in finance literature (e.g. Malmendier and Tate, 2008) that overconfident decision-makers take on significantly higher share of risky (although often unprofitable) projects. Consequently, we control for *overconfidence* estimated as an indicator variable that is equal to 1 if the executive holds exercisable stock options that are more than 67% in the money at least twice over the sample period, and 0 otherwise. We classify the executive as overconfident starting from the first time he/she exhibits such behavior. We estimate the measure of overconfidence following Campbell et al. (2011). In all our regressions, the control variables are measured at the beginning of the period over which volatilities are estimated.

3.2.5 Descriptive Statistics

Table 1 presents descriptive statistics for each of the main variables[11]. The table shows that on average, our sample firms are large, profitable, and moderately leveraged, with total assets of approximately $0.5 billion, return 17.6%, and a leverage ratio of 44%. These sample firms also possess important growth opportunities as implied by an average market-to-book ratio of 2.89. The average CEO is 54 years old. The mean CEO total connectedness is approximately 180, with a median of 93, while the interquartile range equals 129. On average, a CEO is connected with 28 individuals via educational links (interquartile range of 40) and with 146 individuals via employment links (interquartile range of 146). CEOs possess relatively few connections via other social activities, (with a mean of 4 and a median of 1). Overall, these numbers are similar to those reported by prior research (e.g., Cassell et al., 2012; Faleye et al., 2014).

4. Main Results

4.1. The Effect of CEO Social Capital on Aggregate Corporate Risk-taking

[11] We winsorize all variables at 1% in each tale in our main tests to minimize the effects of potential outliers, as suggested by Faleye et al. (2014). Our results are robust to the alternative winsorization at 0.1%.

We first analyze the effect of CEO social capital on corporate risk-taking proxied by return and earnings volatilities. Our first hypothesis predicts that CEO social capital has positive implications for corporate risk-taking. We empirically estimate[12] model (1). The results are presented in Table 2, Panel A. In Columns (1)-(3) *Volatility (Return)* which is estimated as the volatility of monthly stock returns constructed over the window *t*+1 to *t*+3 is the dependent variable. We find that coefficient estimate of CEO social capital is positive and statistically significant (p-values<0.001). In Columns (2) and (3) we add industry (defined by two-digit SIC code) and year fixed effects, respectively. We still find that the coefficient estimates of CEO social capital are positive and significant in all three specifications.

In Columns (4)-(6), we use *Volatility (ROA)* estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+3, as a dependent variable and estimated model (1) without and with industry and year fixed effects. We repeat the same analysis in Columns (7)-(9), but use *Volatility(ROE)* estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+3, as a dependent variable. In each of the alternative models we find that coefficient estimate of CEO social capital is positive and statistically significant (with p-values<0.001). The Coefficient estimates of the control variables are broadly consistent with theory and prior empirical evidence.

Collectively, the results in Table 2 are consistent with our prediction and suggest that there is a positive association between CEO social capital and corporate risk-taking[13]. The effects of social capital are also economically significant. For example, consider specification in Columns

[12] We note that standard errors in all main regressions are adjusted for heteroscedasticity (White, 1980) and clustered by firm to control for serial dependence (Petersen, 2009). In robustness tests we cluster standard errors by CEO, as suggested by Faleye et al. (2014).

[13] We also use five components of CEO social capital (that is, educational, employment, and other connections via social clubs/non-profit organizations as well as social connection with financier firms and with S&P500 firms) discussed above as the main explanatory variables. Our results, not tabulated for brevity, suggest that aggregate risk-taking is increasing in all of the components of the CEO social capital, having employment connections and social capital resident in social networks with S&P 500 firms the strongest effects.

(3), (6) and (9). Increase in CEO social connectedness from the first to the third quartile results in 2.97% (Column 3), 7.47% (Column 6), and 16.59% (Column 9) increase in risk-taking relative to the cross-sectional mean[14].

4.2. CEO Social Capital and Riskiness of Investment and Financial Policies

Next, we investigate the effects of CEO social capital and corporate policies. Our hypotheses predict that CEOs with high social capital adopt riskier investment and financial policies, respectively. We first examine the association between CEO social capital and investment policies as proxied by R&D expenditures[15] and corporate diversification. Prior research (e.g., Kothari et al., 2002; Coles et al., 2006) suggests that R&D development expenditures are more risky than other investment choices. In addition, CEOs could alter riskiness of investment choices by diversifying firms' operations among different industry segments (e.g., Amihud and Lev, 1981; Coles et al., 2006). Consequently, CEO social capital should positively affect R&D expenditure and negatively diversification of firm investments.

We empirically estimate model (1) and use our two measures of riskiness of corporate investment policies as dependent variables. The results are presented in Table 2, Panel B. In Columns (1)-(2) the dependent variable is *R&D/Sales* - the ratio of research and development expenses to total sales, estimated as the average over the window *t*+1 to *t*+3. We find that coefficient estimate of social capital is positive and significant in both models. In Columns (3)-(4) we examine the relation between CEO social capital and *Diversification* - the measure of corporate diversification estimated following the methodology of Jacquemin and Berry (1979), constructed as the average over the window *t*+1 to *t*+3. Consistent with our predictions,

[14] We first estimate percentage change (p) in CEO social connectedness from quartile 1 to quartile 3 and then calculate $\beta \times \log\left[\frac{100+p}{100}\right]$; next we compare it to the cross-sectional mean of Volatility (Return), Volatility (ROA), and Volatility (ROE).

[15] In the pioneering study of CEO social connectedness and innovation, Faleye at al. (2014) is the first to uncover the connectedness-R&D relation.

coefficient estimates of corporate diversification is negative and significant suggesting negative association between CEO social capital and diversification of firm investments.

Next we analyze the relation between CEO social capital and corporate financial policies. We expect that well-connected CEOs will adopt less conservative financial policies. CEOs could increase riskiness of financial policies by holding less liquid assets or by increasing debt burden of the firm (e.g., Begley et al., 1996). Thus, we expect a positive relation between CEO social capital and financial leverage and negative association between CEO social capital and asset liquidity (working capital).

In Columns (5)-(6) of Table 2, Panel B, we use *Leverage* estimated as the ratio of total debt to total assets, measured as the average over the window t+1 to t+3, as the dependent variable[16]. We find that consistent with our prediction, CEO social capital positively affects firm leverage. Columns (7)-(8) present estimation results of model (1) where we use *Working Capital* - current assets minus current liabilities, scaled by total assets, measured as the average over the window t+1 to t+3, as the dependent variable. We discover that the coefficient estimate of CEO social capital is negative and significant, as predicted.

Taken together, the results depicted in Table 2, Panel B demonstrate that consistent with our prediction CEO social capital increases riskiness of firm investment and financial policies. The coefficient estimate of the main independent variable is also economically significant. For example, the leverage of a firm whose CEO is at the third quartile of social capital is 0.025 units higher than that of a firm with CEO in the first quartile, which is 4.90% of the cross-sectional mean of the leverage ratio.

4.3. CEO Social Capital, Corporate Policies, and Risk-taking – Disentangling the Channels

[16] We note that in our financial policy regressions we drop debt/equity ratio as one of the explanatory variables and add the natural logarithm of market value of equity.

In this section we focus on the effects of corporate policies on the relation between CEO social capital and corporate risk-taking. We measure corporate risk-taking by the volatilities of returns and earnings. We argue that CEO social capital causes corporate policy actions and these actions lead to greater volatilities. More specifically, social capital causes increase in the riskiness of investment and financial policies which in turn make future cash flows more uncertain, and consequently firm stock returns and earnings more volatile. To quantify these indirect effects of CEO social capital on return and earning volatilities through investment and financial policies we use the methodology[17] that requires estimating the following three equations (Preacher and Hayes, 2004):

Risk-taking=f(CEO Social Capital, Controls) (2)
Corporate Policies=f(CEO Social Capital, Controls) (3)
Risk-taking=f(CEO Social Capital, Corporate Policies, Controls) (4)

We perform the first step (equation 2) of the analysis in Table 2, Panel A (subsection 4.1) by establishing significant effects of CEO social capital on volatilities of returns and earnings. The results of the second step (equation 3) are discussed in the previous section (Table 2, Panel B) where we show that CEO social capital significantly affects corporate investment and financial policies. The final step is to include corporate policies in the regression of CEO social capital on return and earnings volatilities (equation 4). The main variable of interest is the reduction in the effects of CEO social capital on return and earnings volatilities.

We present the results of this analysis in Table 3. In Panel A, the dependent variable is *Volatility (Return)* which is estimated as the volatility of monthly stock returns constructed over the window t+1 to t+3. In Columns (1)-(4) the policy variables are: *R&D/Sales* - the ratio of

[17] More formally this methodological approach is called the mediation analysis (first introduced by Baron and Kenny, 1986). It is often used in business disciplines (see recent research published in reputable business journals: e.g., Management - Rungtusanatham et al., 2014, Marketing - Feinberg, 2012, Entrepreneurship - Semrau and Sigmund, 2012), as well as in psychology (see Pearl, 2014; Imai et al., 2010). In Finance, see application in Fedaseyeu et al. 2016.

research and development expenses to total sales, *Diversification* - the measure of corporate diversification estimated following the methodology of Jacquemin and Berry (1979), *Leverage* - the ratio of total debt to total assets, and *Working Capital* - current assets minus current liabilities, scaled by total assets, respectively. In Column (5) we include all corporate policies together[18]. Including the measures of corporate policies always reduces the effect of CEO social capital on return volatility. This decrease represents 11.44% for the model with R&D/Sales, 24.22% for the model with Leverage, 1.15% for the model with Diversification, and 1.15% for the model with Working Capital. When all policies are included together, the reduction 31.47%. We observe significant effects of R&D/Sales (p-value<0.001), and Leverage (p-value<0.001). The results with Diversification and Working Capital are marginal.

In Panel B of Table 3 the dependent variables are in Columns (1)-(5) *Volatility (ROA)* which is estimated as the volatility of a firm's quarterly return on asset, and in Columns (6)-(10) *Volatility (ROE)* which is estimated as the volatility of a firm's quarterly return on asset. In Columns (1)-(4) (Columns 6-9) the policy variables are: *R&D/Sales*, *Diversification*, *Leverage*, and *Working Capital*, respectively. In Column 5 (Column 10) we include all corporate policies together. We observe a significant channeling effects in all specifications (but Columns (2) and (7) where the results are marginal). We find the strongest effects through R&D/Sales and Leverage: 20.27% and 33.75% in the model with Volatility (ROA), and 16.67% and 85.20% in the models with Volatility (ROE), respectively. The models with multiple policies imply nearly full channeling (p-values of the effects are less than 0.001). The coefficient estimate of CEO

[18] We note that to estimate the channeling effects of financial policies we first re-estimated our model (2) with dependent variables Volatility (Return), Volatility (ROA), and Volatility (ROE), but drop debt/equity ratio as one of the explanatory variables and add the natural logarithm of market value of equity. This resulted in the coefficient estimate of social capital 0.00607 for the model with Volatility (Return), 0.0080 – for the model with Volatility (ROA), and 0.02318 for the model with Volatility (ROE). Next we use these coefficient estimates to calculate channeling effects in Panel A, Columns (3)-(5) and in Panel B, Columns (3)-(5) and (8)-(10).

social capital is insignificant (Columns 5 and 10). However, statistical insignificance does not necessarily imply full mediation. Consequently, we look at the actual decrease of the coefficient estimates of the main independent variables and document 67.5% and 97.97% reduction (in Columns 5 and 10, respectively).

Overall, the results in this section present strong evidence that the effects of CEO social capital on return and earnings volatilities are channeled by corporate policies. However, controlling for corporate policies does not eliminate the effect of CEO social capital on volatilities in some models. This (remaining) direct effect is consistent with the following explanation: our four measures of corporate policies likely do not capture all aspects of riskiness of corporate investment and financial policies (such as risky M&A activities).

4.4. CEO Social Capital, Corporate Risk-taking: Second-Order Moderating Effects

In this section, we examine the moderating effects of managerial entrenchment, free cash flow, external financing, information asymmetry, and investment efficiency on the implications of CEO social capital for corporate risk-taking. We partition our sample using respective median values of moderating variables and estimate equation (1) for each of the subsamples.

Table 4 depicts the results of the split-sample analysis. The dependent variable is *Volatility (Return)* - volatility of monthly stock returns[19]. Specifically, in columns (1)-(2) we use *CEO Pay Slice* as a conditioning variable. The coefficient estimate for CEO social capital is 2.2 times higher in the subsample of firms with low CEO pay slice (with p-value of the difference 0.0035). Columns (3)-(4) show that social capital is stronger (2.5 times) predictor of risk-taking for firms with low free cash flow. Further, Columns (5)-(6) document that the effect of social capital on risk-taking is stronger for firms with high external financing. Regressions in columns (7)-(8) are

[19] We note that the results of the split-sample analysis are qualitatively similar if *Volatility (ROA)* or *Volatility (ROE)* is used as the dependent variable.

estimated for the high and low information asymmetry subsamples. We find that in the low information asymmetry subsample, the implications of social capital on risk-taking are marginally stronger (p-value 0.1243). Finally, columns (9)-(10) show that social capital exerts stronger influence (1.53 times) on risk-taking for firms that previously underinvest.

In summary, the results in this section provide strong supportive evidence to our conjectures that the effects of social capital on risk-taking should be stronger for firms with low managerial entrenchment, low free cash flows, high external equity financing, high information asymmetry, and firms that previously underinvest.

5. Robustness Analysis

In this section, we conduct several tests to examine the robustness of our main findings. Our measure of CEO social capital could be correlated with CEO human capital. Specifically, more skilled and intelligent individuals may have easier time networking (e.g. more people want to be connected to such individuals). Consequently, more intelligent/skilled CEOs might take advantage of risky growth opportunities and thus the effects of CEO social capital on risk-taking could be due to superior CEO human capital (and not social capital). We address this possibility by filtering the human capital out of CEO social capital measure. That is, we use excess connectedness - $SC(Total)_{excess}$, estimated as the residuals from the regression of social capital on CEO human capital index, as the alternative measure of social capital. Next, we employ measures of aggregate risk-taking as well as corporate policies estimated over the window t+1 to t+5 to examine whether our results are robust to the alternative estimation period.

In addition, we use the alternative model specification to test whether our results are dependent on the specific controls used in our main tests. Particularly, we add two firm-level variables: *Asset Tangibility* estimated as the ratio of fixed assets (property, plant and equipment) to total assets, *and Free Cash Flow*, estimated as the cash flow from operations minus cash dividends, scaled by lagged total assets, to account for the potential of financial distress and availability of funds to invest in new projects.

Finally, we also add additional CEO-level controls: natural logarithm of the sum of salary and bonus compensation as to proxy for the CEO outside wealth, and two other variables to measure incentives arising from equity-based compensation, *Delta* - the sensitivity of the value of the CEO's accumulated equity-based compensation to a one-percent change in the stock price, and *Vega* - the sensitivity of the value of the CEO's accumulated equity-based compensation to a one-percent change in the volatility of stock prices as controls (Coles et al., 2006). We also include CEO *Overconfidence* as an additional control to account for the possibility that overconfident CEOs engage in excessive (although often value-destroying) risk-taking.

In Table 5 we estimate our model using the alternative measure of social capital ($SC(Total)_{excess}$) and additional controls. In Panel A, the dependent variable in Columns (1)-(3) is *Volatility (Return)* which is estimated as the volatility of monthly stock returns constructed over the window t+1 to t+5. In Columns (4)-(6), the dependent variable is *Volatility (ROA)* which is estimated as the volatility of a firm's quarterly return on asset, constructed over the window t+1 to t+5. In Columns (7)-(9) the dependent variable is *Volatility (ROE)* which is estimated as the volatility of a firm's quarterly return on asset, constructed over the window t+1 to t+5. We find that coefficient estimate of CEO social capital remains positive and highly significant. Thus, we conclude that the effects documented in the main tests are robust to the additional controls and alternative variable measurements.

Next we check robustness of our findings that CEO social capital induces more risky investment and financial policies. In Panel B of Table 5 we re-estimate our corporate policy measures over the window t+1 to t+5 and add additional controls in our regressions. We find that the coefficient estimates of the riskiness of corporate investment and financing policy choices have predicted signs and remain significant.

In addition, we examine whether our finding that corporate policy choices channel the effects of CEO social capital on volatility in stock returns and in earnings, is robust to the alternative measures of CEO social capital, other variables, as well as model specification. In Table 6, Panel A, we use *Volatility (Return)*, estimated as the volatility of monthly stock returns constructed over the window t+1 to t+5. Inclusion of corporate investment and financing policy variables always reduces the effect of CEO social capital on return volatility[20]. In Panel B we use in Columns (1)-(5) *Volatility (ROA)*, estimated as the volatility of a firm's quarterly return on asset, and in Columns (6)-(10) *Volatility (ROE)*, estimated as the volatility of a firm's quarterly return on asset. Both variables are measured over the window t+1 to t+5. We find the convincing evidence of mediation. Specifically, when all policy measures are included, the coefficient estimate of social capital drops nearly by 56% in the model with the *Volatility (ROA)* and almost by 89% - in the models with *Volatility (ROE)*.

Finally, we also confirm the robustness of our findings that CEO social capital effects are moderated by managerial entrenchment and other variables[21]. In Table 7, Panel A, we use the following proxies for managerial entrenchment: *GIM Index* from Gompers et al. (2003), *BCF Index* from Bebchuk et al. (2009), *Co-option* from Coles et al. (2014), and *Hostile Takeover*

[20] We note that as we clarified in footnote 18, to estimate the channeling effects of financial policies we first re-estimated our model (2) with dependent variables Volatility (Return), Volatility (ROA), and Volatility (ROE), but drop debt/equity ratio as one of the explanatory variables and add the natural logarithm of market value of equity. This resulted in the coefficient estimate of social capital 0.0049 for the model with Volatility (Return), 0.0005 – for the model with Volatility (ROA), and 0.0235 for the model with Volatility (ROE). Next we use these coefficient estimates to calculate channeling effects in Panel A, Columns (3)-(5) and in Panel B, Columns (3)-(5) and (8)-(10).

[21] In Table 7 we use our main aggregate risk-takin measure - Volatility(Return) - as the dependent variable. The results of the split-sample analysis are qualitatively similar if Volatility (ROA) or Volatility (ROE) is used as the dependent variable. These results are untabulated for brevity, but available upon request.

Threat index from Cain et al. (2017). Consistent with the prediction, we find that social capital effects are significant only in the subsample of firms with low managerial entrenchment. In Panel B of Table 7, we partition the sample based on median values of Free Cash Flow, External Financing, Information Asymmetry and Investment Efficiency. Using excess social capital and alternative model specification, we provide supportive evidence of our main findings. Specifically, we document that social capital exerts significant influence on risk-taking for firms with low free cash flow, high external financing, high information asymmetry, and firms that previously underinvest.

6. Endogeneity Concerns

The results presented above suggest that CEO social capital positively affects the volatility of future stock returns and earnings as well as the riskiness of corporate policies. However, these documented relations might be contaminated by the potential endogeneity.

The primary concern is whether the firms that take more risk could employee well connected CEOs, raising the possibility that CEO social capital and various dependent variables are jointly determined. To address this possibility, we employ two-stage least squares (2SLS) framework. Prior research employing the potential instruments for social networks is relatively scant, so finding the appropriate instrument can be challenging. The basic requirement for validity of the instrument is that the instrument should have no effect on the dependent variable other than through its effect on the suspected endogenous variable.

We deploy two alternative instrumental variables. First, we posit that the number of local social organizations is likely to influence CEO social capital as measured by social connectedness but there is no reason to believe that it could have implications for return and earnings volatilities and corporate policies. Therefore, this instrument is reasonably exogenous.

Second, we use average connectedness of CEOs within 100 miles of the firm's headquarters (excluding the firm (and consequently the CEO) in question) as an instrument. This variable can be a valid instrument because when the location fixed-effects are present, the location-mean CEO social connectedness is not likely to affect our dependent variables. On the other hand, it would certainly impact social connections of a particular CEO.

We report the results from the first and the second stage regressions in Table 8. Column (1) reports the results from the first stage estimations. As expected, the our instruments are highly correlated with the CEO social capital. Kleibergen-Paap statistics is rejecting the null hypothesis that the instruments are weak (with p-value<0.0001). In addition, F-test of excluded instruments is greater than 10. The second-stage regression results confirm our previous findings of a statistically and economically significant relationship between the CEO social capital and the volatilities of future stock returns and earnings, as well as the riskiness of corporate policies[22].

A related concern is that our results are driven by reverse causality. It is plausible that the CEO could pursue risky activities at time t with the expectation to enhance his/her social network at t+1 or t+2 possibly for expecting a salary increase (rolodex effect from Engelberg et al., 2013). This could generate the reverse causality. We test this possibility by examining the time-series evolution of CEO social capital. Specifically, we regress the annual percentage change in CEO social connectedness constructed over the various windows (t to t+1, t+1 to t+2, t+2 to t+3, and t to t+3) on lagged measures of risk-taking, corporate policies, and controls. We present the results of this analysis in Appendix A documenting that risk-taking proxies cannot explain time-series changes in social connectedness[23].

[22] The number of industries in which the CEO has worked in the past could also be valid instrument for CEO social capital (Faleye et al. 2014). Our results, not tabulated for brevity, are robust to this alternative instrumental variable.
[23] We note that we further examine time-series properties of our main effects by looking at CEO turnover. Specifically, we identify subsample of firms where a new CEO has significantly higher connectedness than the

Finally, skeptical readers might contend that firm fixed-effect regressions are necessary to control for unobservable time-invariant characteristics that may affect corporate risk-taking. However, fixed-effects models rely exclusively on within-subject variations to identify the effects of independent variables, and our CEO social capital measures do not vary much from year to year in our sample. Consequently, we argue that the inclusion of CEO and firm fixed effects is not appropriate in our research design because we have a short sample period and our social capital measures are relatively constant. High standard deviations of social capital measures are almost entirely due to cross-sectional variation. The median one year change in (non-log) CEO social connectedness measure is 3, compared to the average connections of 179. Thus, including fixed effects makes identification of network effects almost impossible.

7. CEO Social Capital and Firm Value

In this section we examine the link between CEO social capital, corporate risk-taking, and firm value. If social capital encourages managerial risk-taking that is optimal for shareholders, we predict a positive relation between increased risk-taking attributable to CEO social capital and firm value. However, higher risk due to higher connectedness may not unconditionally result in higher value (e.g., El-Khatib et al., 2015).

Following methodology developed by prior research (e.g., Bowen et al., 2006; Jiraporn and Liu, 2008; Core et al., 1999) we estimate "excess risk-taking" that is beyond the usual risk-taking determined by standard economic factors. More specifically, using measures of aggregate risk-

predecessor (requiring non-missing observation for CEO/Firm variables for at least three years before and after turnover). Results, not tabulated for brevity, show that appointment of a new CEO with significantly higher connectedness than the predecessor is associate with a significant increase in firm risk (specifically, the average increase in return volatility is 0.01 with p-value of 0.0508). In addition, we also look at whether the likelihood of appointing the highly-connected CEO is higher when the firm currently under-invests. The results (unreported) show that the probability of appointing the highly-connected CEO (defined as an indicator variable that equals 1 if CEO social capital is in the top tercile) is significantly higher for firms that currently under-invest.

taking as the dependent variables, we estimate two regressions, one with all the independent variables, and the other with every variable except CEO social capital. The difference in the predicted values of risk-taking from the two regressions is "excess risk-taking" directly attributed to the CEO social capital. Next we estimate the following regression:

Abnormal Valuation =f(Excess risk-taking, Controls) (5)

Abnormal Valuation is measured by risk-adjusted abnormal stock return defined as the excess one year (three-year) holding period return, which equals the difference between the actual return and expected return estimated using Fama-French four factor model[24]. The regression results are shown in Table 9. The major independent variables of interest are "excess risk-takings" measured by *Abnormal Volatility*, constructed using *Volatility(Return)* - volatility of monthly stock returnsover the window *t*-1 to *t*-3. The sample is partitioned based on respective median values of CEO Pay Slice (Columns 1-2), Free Cash Flow (Columns 3-4), External Financing (Columns 5-6), Information Asymmetry (Columns 7-8), and Investment Efficiency (Columns 9-10). The coefficient of predicted "excess risk-taking" (abnormal volatility) is positive and significant[25] for firms with low managerial entrenchment, high external equity financing, high information asymmetry, and firms that previously underinvest[26] (results for high vs. low free cash flow are marginal). In summary, the results in this section indicate that social capital induces CEOs to take more risk which is value-enhancing conditional on a set of moderating variables.

[24] Specifically, for this estimation we regress the past 24 months of stock returns on returns of risk factors to estimate the firm's risk exposure (betas) for each of the four factors. Next, we calculate the expected holding period return on the stock over the next 12 (36) months using the estimated betas and the expected returns of risk factors (estimated as the long-term averages of excess returns for market, size, market-to-book, and momentum factors). The abnormal risk-adjusted holding period return is the difference between the actual return over the next 12 (36) months and the expected holding period return.

[25] For the reporting convenience, we divide the coefficient estimate of excess risk-taking by 100.

[26] We note that our results are qualitatively similar of using the abnormal risk-adjusted holding period return over the next 36 months as a dependent variable. The results are also qualitatively unchanged if Volatility (ROA) or Volatility (ROE) is used for measuring Abnormal Volatility.

8. Conclusion

Understanding the effects of CEO personal characteristics on corporate risk-taking is important because the CEO, as a firm's chief agent, sets the tone for the riskiness of corporate policies. A number of classical studies show that managers prefer "quiet life" and tend to be more risk-averse than shareholders would like them to be. Recent research emphasizes the importance of CEO personal attributes for explaining corporate risk-taking choices. In this study, we focus on the role of previously unexamined factor, CEO social capital, in ameliorating risk aversion incentives.

We propose several arguments why CEO social capital could positively affect corporate risk-taking. Social capital offers a way to pool individual risks by providing informal insurance within a social network and thus stimulates riskier preferences. Further, social capital intensifies a manager's sense of power, leading to higher managerial risk-tolerance and ultimately greater corporate risk-taking. Finally, a potential loss of employment is perhaps the most important cause for sub-optimal risk-taking. CEO social capital can provide a safety net that enhances the likelihood of re-employment should the CEO lose his job.

We test our hypothesis on a sample of US firms and find that CEO social capital positively affects corporate risk-taking choices. We further show that social ties cause corporate policy actions, and that these actions lead to greater volatilities in earnings or in stock returns. We also uncover several important second-order magnifying (mitigating) factors that significantly moderate the effects of social capital on corporate risk-taking. In addition, we find that the increase in risk taking attributable to CEO social capital has implications for risk-adjusted valuation. These results are robust to alternative model specifications, alternative variable measurement, and tests for an endogeneity.

Our findings have important implications for the corporate finance literature as our results suggest that social capital of senior executives can be a mechanism for reducing agency problems within a firm by encouraging better alignment of managers’ interests with those of shareholders. In addition, our results offer important insights regarding CEO behaviors and how they shape corporate decisions about risk and project investment.

Appendix: Evolution of CEO Connectedness - Reverse Causality Concerns

This table reports estimates of coefficients using the OLS. The dependent variable in Panel A, Columns (1)-(5) is $\Delta SC(Total)_{t+1}$ - the annual percentage change in *SC(Total)*, constructed over the window *t*, to *t*+1 and in Columns (6)-(10) $\Delta SC(Total)_{t+2}$ - the annual percentage change in *SC(Total)*, constructed over the window *t+1* to *t*+2. The dependent variable in Panel B, Columns (1)-(5) is $\Delta SC(Total)_{t+3}$ - the annual percentage change in *SC(Total)*, constructed over the window *t+2* to *t*+3, and in Columns (6)-(10) $\Delta SC(Total)_{t+1,\, t+3}$ - the percentage change in *SC(Total)*, constructed over the window *t* to *t*+3. *Volatility (Return)* is estimated as the volatility of monthly stock returns constructed over the window *t*-1 to *t*-3. *SC(Total)* is the natural logarithm of 1 plus number of individuals with whom the CEO is connected via educational , employment, or other social links in BoardEx universe. Educational links are estimated as the number of individuals with whom the CEO attended the same university, graduated within 2 years of each other, and earned a similar type of degree. Employment links are constructed as the number of individuals with whom the CEO shares a common employment history. Other social links are defined as the number of individuals with whom the CEO is connected through social clubs or other non-profit organizations. *R&D/Sales Ratio* is the ratio of research and development expenses to total sales, estimated as the average over the window *t*-1 to *t*-3. *Diversification* is the measure of corporate diversification estimated following the methodology of Jacquemin and Berry (1979), constructed as the average over the window *t*-1 to *t*-3. *Leverage* is the ratio of total debt to total assets, measured as the average over the window *t*-1 to *t*-3. *Working Capital* is current assets minus current liabilities, scaled by total assets, measured as the average over the window *t*-1 to *t*-3. Control variables include: *Log(Age)* - the natural logarithm of the age of the firm, *Log(Size)* - the natural log of total assets, *MB Ratio* - the ratio of the market value of equity to the book value of assets, *Sales Growth* - the ratio of total sales to lagged total sales, *Return* - the stock return over fiscal year t, *Debt/Equity Ratio* - the ratio of total debt to the market value of equity, *Cash Surplus* - estimated as net cash flow from operations minus depreciation expense plus research and development expenditures, scaled by total assets, *Log(CEO Age)* - natural log of the CEO age at fiscal year t, and *Log(CEO Tenure)* - the natural log of CEO tenure in years. CEO tenure in a given year is determined as the length of time between the date that the person became the CEO and the current fiscal year end. *CEOGD* is CEO gender dummy variable that equals 1 for male CEOs and 0 otherwise. Robust standard errors, adjusted for heteroskedasticity and clustered at the firm level, are reported in brackets below the coefficients. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: Annual Change in CEO Social Capital (t+1, t+2)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	ΔSC(Total)$_{t+1}$					ΔSC(Total)$_{t+2}$				
Volatility (Return)	-0.0373					-0.0038				
	(0.04440)					(0.05740)				
R&D Sales Ratio		-0.0036					-0.0029			
		(0.00243)					(0.00307)			
Diversification			0.0186					0.0224		

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
			(0.01384)					(0.01708)		
Leverage				-0.0180					-0.0186	
				(0.01712)					(0.02143)	
Working Capital					-0.0169					-0.0060
					(0.02333)					(0.03225)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
adj. R-sq	0.009	0.009	0.009	0.009	0.009	0.010	0.010	0.010	0.010	0.010
N	11220	11220	11220	11220	11220	8485	8485	8485	8485	8485

Panel B: Annual Change in CEO Social Capital (t+1, t+3)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	ΔSC(Total)$_{t+3}$					ΔSC(Total)$_{t+1, t+3}$				
Volatility (Return)	0.2040					0.5786				

	(0.16146)					(0.56208)				
R&D Sales Ratio		-0.0019					-0.0298			
		(0.00403)					(0.02279)			
Diversification			0.0209					0.0748		
			(0.02247)					(0.08492)		
Leverage				-0.0239					-0.2446	
				(0.03082)					(0.23047)	
Working Capital					0.0084					0.0851
					(0.04826)					(0.22932)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
adj. R-sq	0.011	0.010	0.011	0.010	0.010	0.029	0.029	0.029	0.028	0.028
N	6249	6249	6249	6249	6249	6195	6195	6195	6195	6195

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Table 1: Sample Descriptive Statistics

This table reports summary statistics for the main dependent and independent variables. *Connections(Total), Connections(Educ), Connections(Empl), Connections(Other), Connections (Fin)*, and *Connections (S&P)* are the measures of social connections. *Connections(Total)* is the number of individuals with whom the CEO shares a common educational, employment, or social history in BoardEx. *Connections(Educ)* is the number of individuals with whom the CEO attended the same university, graduated within 2 years of each other, and earned a similar type of degree. *Connections(Empl)* is the number of individuals with whom the CEO shares a common employment history. *Connections(Other)* is the number of individuals with whom the CEO is connected through social clubs or other non-profit organizations. *Volatility(Return)* is volatility of monthly stock returns constructed over the window *t*+1 to *t*+3. *Volatility (ROA)* is the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+3, where return on asset is the ratio of operating income to total assets. *Volatility (ROE)* is the volatility of a firm's quarterly return on equity, constructed over the window *t*+1 to *t*+3, where return on equity is the ratio of operating income to total stockholders' equity. *R&D/Sales* is the ratio of research and development expenses to total sales, estimated as the average over the window *t*+1 to *t*+3. *Diversification* is the measure of corporate diversification estimated following the methodology of Jacquemin and Berry (1979), measured as the average over the window *t*+1 to *t*+3. *Leverage* is the ratio of total debt to total assets, measured as the average over the window *t*+1 to *t*+3. *Working Capital* is current assets minus current liabilities, scaled by total assets, measured as the average over the window *t*+1 to *t*+3. *Log(Age)* is the natural logarithm of the age of the firm. *Log(Size)* is the natural log of total assets. *MB Ratio* is the ratio of the market value of equity to the book value of assets. *Sales Growth* is the ratio of total sales to lagged total sales. *Return* is the stock return over fiscal year t. *Debt/Equity Ratio* is the ratio of total debt to the market value of equity. *Cash Surplus* is estimated as net cash flow from operations minus depreciation expense plus research and development expenditures, scaled by total assets. *Log(CEO Age)* is natural log of the CEO age at fiscal year *t*. *Log(CEO Tenure)* is the natural log of CEO tenure in years. CEO tenure in a given year is determined as the length of time between the date that the person became the CEO and the current fiscal year end. *CEOGD* is CEO gender dummy variable that equals 1 for male CEOs and 0 otherwise.

	N	Mean	STD	25%	Median	75%
Connections(Total)	29748	179.015	225.285	37	93	222
Connections(Educ)	29748	28.453	39.597	1	11	41
Connections(Empl)	29748	146.412	205.425	24	62	170
Connections(Other)	29748	4.291	21.283	1	1	1
Volatility (Return)	29748	0.141	0.075	0.088	0.123	0.172
Volatility (ROA)	29748	0.019	0.021	0.006	0.012	0.022
Volatility (ROE)	29748	0.115	0.340	0.014	0.026	0.054
R&D/Sales	29748	0.208	0.949	0.000	0.003	0.075
Diversification	29748	0.172	0.320	0.000	0.000	0.216
Leverage	29748	0.504	0.250	0.317	0.489	0.650
Working Capital	29748	0.262	0.231	0.092	0.240	0.419
Log(Age)	29748	2.757	0.740	2.197	2.708	3.367

Log(Size)	29748	6.197	2.047	4.733	6.110	7.536
MB Ratio	29748	2.894	3.862	1.214	2.024	3.496
Sales Growth	29748	1.153	0.436	0.978	1.083	1.215
Return	29748	0.176	0.707	-0.231	0.057	0.377
Debt/Equity Ratio	29748	0.444	0.948	0.006	0.132	0.416
Cash Surplus	29748	0.064	0.125	0.011	0.064	0.126
Log(CEO Age)	29748	3.992	0.149	3.892	4.007	4.094
Log(CEO Tenure)	29748	1.346	0.889	0.693	1.386	1.946
CEOGD	29748	0.975	0.157	1.000	1.000	1.000

Table 2: CEO Social Capital, Risk-taking and Corporate Policies

This table reports estimates of coefficients using the OLS. In Panel A, the dependent variables are: in Columns (1)-(3) *Volatility (Return)* which is estimated as the volatility of monthly stock returns constructed over the window *t*+1 to *t*+3, in Columns (4)-(6) *Volatility (ROA)* - estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+3, and in Columns (7)-(9) *Volatility (ROE)* - estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+3. Return on asset is the ratio of operating income to total assets and return on equity is the ratio of operating income to total stockholders' equity. In Panel B, the dependent variables are: in Columns (1)-(2) *R&D/Sales* - the ratio of research and development expenses to total sales, estimated as the average over the window *t*+1 to *t*+3; in Columns (3)-(4) *Diversification* - the measure of corporate diversification estimated following the methodology of Jacquemin and Berry (1979), constructed as the average over the window *t*+1 to *t*+3; in Columns (5)-(6) *Leverage* - the ratio of total debt to total assets, measured as the average over the window *t*+1 to *t*+3; in Columns (7)-(8) *Working Capital* - current assets minus current liabilities, scaled by total assets, measured as the average over the window *t*+1 to *t*+3. *SC(Total)* is the natural logarithm of 1 plus number of individuals with whom the CEO is connected via educational, employment, or other social links in BoardEx universe. Educational links are estimated as the number of individuals with whom the CEO attended the same university, graduated within 2 years of each other, and earned a similar type of degree. Employment links are constructed as the number of individuals with whom the CEO shares a common employment history. Other social links are defined as the number of individuals with whom the CEO is connected through social clubs or other non-profit organizations. Control variables are defined in the Table 1 header. Robust standard errors, adjusted for heteroskedasticity and clustered at the firm level, are reported in brackets below the coefficients. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: CEO Social Capital and Aggregate Corporate Risk-taking

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Corporate Risk-taking Measures								
	Volatility (Return)			Volatility (ROA)			Volatility (ROE)		
SC(Total)	0.0049***	0.0039***	0.00526***	0.0018***	0.0018***	0.00178***	0.0285***	0.0260***	0.02394***
	(0.00068)	(0.00068)	(0.00068)	(0.00024)	(0.00023)	(0.00023)	(0.00356)	(0.00335)	(0.00345)
Log (Age)	-0.0162***	-0.0171***	-0.0159***	-0.0018***	-0.0020***	-0.0022***	-0.0135**	-0.0149**	-0.0187***
	(0.00112)	(0.00110)	(0.00109)	(0.00034)	(0.00033)	(0.00034)	(0.00599)	(0.00591)	(0.00610)
Log (Size)	-0.0146***	-0.0137***	-0.0142***	-0.0042***	-0.0044***	-0.0044***	-0.0216***	-0.0209***	-0.0207***
	(0.00047)	(0.00049)	(0.00048)	(0.00018)	(0.00018)	(0.00018)	(0.00254)	(0.00257)	(0.00256)

MB Ratio	0.0007***	0.0007***	0.0004**	0.0006***	0.0005***	0.0005***	0.0098***	0.0083***	0.0084***
	(0.00016)	(0.00015)	(0.00014)	(0.00007)	(0.00007)	(0.00007)	(0.00184)	(0.00183)	(0.00184)
Sales Growth	0.0106***	0.0091***	0.0065***	0.0047***	0.0034***	0.0033***	0.0396***	0.0289***	0.0294***
	(0.00122)	(0.00123)	(0.00122)	(0.00053)	(0.00049)	(0.00050)	(0.00890)	(0.00849)	(0.00871)
Return	-0.0048***	-0.0049***	-0.0006	-0.0016***	-0.0016***	-0.0014***	-0.0173***	-0.0149***	-0.0158***
	(0.00065)	(0.00063)	(0.00061)	(0.00019)	(0.00018)	(0.00019)	(0.00376)	(0.00365)	(0.00393)
Debt/Equity Ratio	0.0171***	0.0182***	0.0179***	-0.0012***	-0.0010***	-0.0009***	0.0595***	0.0625***	0.0645***
	(0.00105)	(0.00103)	(0.00102)	(0.00019)	(0.00019)	(0.00019)	(0.00710)	(0.00720)	(0.00729)
Cash Surplus	-0.1043***	-0.1113***	-0.1163***	-0.0215***	-0.0212***	-0.0218***	-0.2759***	-0.2663***	-0.2719***
	(0.00553)	(0.00543)	(0.00540)	(0.00262)	(0.00260)	(0.00260)	(0.04477)	(0.04546)	(0.04565)
Log (CEO Age)	-0.0226***	-0.0235***	-0.0211***	0.0029*	0.0013	0.0011	0.0216	0.0148	0.0068
	(0.00484)	(0.00459)	(0.00454)	(0.00164)	(0.00153)	(0.00154)	(0.02341)	(0.02344)	(0.02353)
Log (CEO Tenure)	-0.0031***	-0.0030***	-0.0025***	-0.0008***	-0.0009***	-0.0009***	-0.0115***	-0.0122***	-0.0124***
	(0.00070)	(0.00066)	(0.00064)	(0.00023)	(0.00021)	(0.00021)	(0.00346)	(0.00343)	(0.00342)
CEOGD	0.0005	-0.0014	-0.0019	-0.0029*	-0.0024	-0.0023	-0.0228	-0.0086	-0.0058
	(0.00413)	(0.00394)	(0.00389)	(0.00171)	(0.00156)	(0.00156)	(0.02813)	(0.02744)	(0.02752)
Constant	0.3335***	0.3114***	0.3356***	0.0295***	0.0425***	0.0447***	0.0291	0.0071	0.0288
	(0.01896)	(0.02204)	(0.02223)	(0.00649)	(0.00876)	(0.00881)	(0.09559)	(0.09780)	(0.09907)

Industry FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Year FE	No	No	Yes	No	No	Yes	No	No	Yes
Adj. R-sq	0.302	0.338	0.433	0.225	0.306	0.309	0.068	0.101	0.103
N	29748	29748	29748	29748	29748	29748	29748	29748	29748

Panel B: CEO Social Capital and Corporate Policies

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Corporate Policies							
	R&D/Sales		Diversification		Leverage		Working Capital	
SC(Total)	0.0611***	0.0594***	-0.0132***	-0.0071*	0.0308***	0.0317***	-0.0055**	-0.0075***
	(0.01065)	(0.01098)	(0.00409)	(0.00419)	(0.00305)	(0.00311)	(0.00255)	(0.00262)
Log (Age)	-0.1052***	-0.1066***	0.1077***	0.1173***	0.0230***	0.0243***	-0.0218***	-0.0247***
	(0.01683)	(0.01695)	(0.00831)	(0.00863)	(0.00529)	(0.00543)	(0.00434)	(0.00445)
Log (Size)	-0.0671***	-0.0665***	0.0288***	0.0277***	0.0111***	0.0108***	-0.0272***	-0.0268***
	(0.00779)	(0.00778)	(0.00365)	(0.00364)	(0.00214)	(0.00216)	(0.00184)	(0.00186)
MB Ratio	0.0155***	0.0159***	-0.0029***	-0.0035***	-0.0007	-0.0008	0.0027***	0.0029***
	(0.00365)	(0.00367)	(0.00074)	(0.00075)	(0.00095)	(0.00096)	(0.00069)	(0.00069)

Sales Growth	0.0743**	0.0786**	0.0083**	0.0074*	-0.0060	-0.0067	0.0087**	0.0090***
	(0.02941)	(0.03055)	(0.00408)	(0.00418)	(0.00480)	(0.00492)	(0.00340)	(0.00349)
Return	-0.0333***	-0.0329***	0.0113***	0.0100***	-0.0083***	-0.0085***	0.0115***	0.0128***
	(0.00939)	(0.01060)	(0.00224)	(0.00245)	(0.00211)	(0.00233)	(0.00161)	(0.00178)
Debt/Equity Ratio	-0.0336***	-0.0336***	-0.0061	-0.0083*				
	(0.00681)	(0.00694)	(0.00491)	(0.00499)				
Cash Surplus	-0.8194***	-0.8212***	-0.2087***	-0.1923***	-0.3934***	-0.3915***	0.2236***	0.2173***
	(0.12817)	(0.12901)	(0.02219)	(0.02206)	(0.02988)	(0.02995)	(0.02290)	(0.02291)
Log (CEO Age)	-0.0029	-0.0065	0.0606**	0.0811***	0.0117	0.0139	0.0026	-0.0037
	(0.08013)	(0.07997)	(0.02731)	(0.02738)	(0.02087)	(0.02088)	(0.01857)	(0.01858)
Log (CEO Tenure)	0.0044	0.0042	0.0036	0.0046	-0.0111***	-0.0110***	0.0094***	0.0091***
	(0.00915)	(0.00920)	(0.00400)	(0.00397)	(0.00290)	(0.00290)	(0.00249)	(0.00249)
CEOGD	0.0631	0.0643	0.0122	0.0051	0.0403**	0.0393**	-0.0391**	-0.0368**
	(0.05592)	(0.05594)	(0.03076)	(0.03080)	(0.01953)	(0.01958)	(0.01678)	(0.01684)
Constant	0.3800	0.3394	-0.0053	-0.0447	0.0992	0.1160	0.5231***	0.5230***
	(0.32370)	(0.32254)	(0.18626)	(0.18436)	(0.10544)	(0.10583)	(0.08030)	(0.08038)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Year FE	No	Yes	No	Yes	No	Yes	No	Yes
Adj. R-sq	0.241	0.241	0.192	0.206	0.240	0.241	0.402	0.405
N	29748	29748	29748	29748	29748	29748	29748	29748

Table 3: CEO Social Capital, Corporate Risk-taking, and Corporate Policies – Disentangling the Channels

This table reports estimates of coefficients from an OLS regression. The dependent variable in Panel A is *Volatility (Return)* which is estimated as the volatility of monthly stock returns constructed over the window *t*+1 to *t*+3. In Panel B the dependent variables are in Columns (1)-(5) *Volatility (ROA)* which is estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+3, and in Columns (6)-(10) *Volatility (ROE)* which is estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+3. Return on asset is the ratio of operating income to total assets and return on equity is the ratio of operating income to total stockholders' equity. *SC(Total)* is the natural logarithm of 1 plus number of individuals with whom the CEO is connected via educational, employment, or other social links in BoardEx universe. Educational links are estimated as the number of individuals with whom the CEO attended the same university, graduated within 2 years of each other, and earned a similar type of degree. Employment links are constructed as the number of individuals with whom the CEO shares a common employment history. Other social links are defined as the number of individuals with whom the CEO is connected through social clubs or other non-profit organizations. *R&D/Sales* is the ratio of research and development expenses to total sales, estimated as the average over the window *t*+1 to *t*+3. *Diversification* is the measure of corporate diversification estimated following the methodology of Jacquemin and Berry (1979), constructed as the average over the window *t*+1 to *t*+3. *Leverage* is the ratio of total debt to total assets, measured as the average over the window *t*+1 to *t*+3. *Working Capital* is current assets minus current liabilities, scaled by total assets, measured as the average over the window *t*+1 to *t*+3. *Log(Age)* is the natural logarithm of the age of the firm. Control variables are defined in the Table 1 header. The channeling effect is the reduction in the effects of CEO social capital on return and earnings volatilities, estimated as the change in the coefficient estimates of social capital from the respective values reported in Panel A of Table 2. Robust standard errors, adjusted for heteroskedasticity and clustered at the firm level, are reported in brackets below the coefficients. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: Dependent variable - Volatility (Return)

	(1)	(2)	(3)	(4)	(5)
SC(Total)	0.00472***	0.0052***	0.0046***	0.0060***	0.00416***
	(0.00067)	(0.00068)	(0.00067)	(0.00068)	(0.00065)
R&D/Sales	0.0092***				0.0092***
	(0.00087)				(0.00095)
Diversification		-0.0086***			-0.0105***
		(0.00222)			(0.00220)
Leverage			0.0465***		0.0548***
			(0.00358)		(0.00401)
Working Capital				-0.0092**	0.0165***
				(0.00436)	(0.00484)

Log (Age)	-0.0149***	-0.0149***	-0.0185***	-0.0176***	-0.0157***
	(0.00108)	(0.00113)	(0.00109)	(0.00112)	(0.00112)
Log (Size)	-0.0136***	-0.0140***	-0.0148***	-0.0146***	-0.0140***
	(0.00048)	(0.00049)	(0.00046)	(0.00048)	(0.00047)
MB Ratio	0.0002	0.0003**	0.0012***	0.0012***	0.0009***
	(0.00014)	(0.00014)	(0.00015)	(0.00015)	(0.00015)
Sales Growth	0.0057***	0.0065***	0.0081***	0.0079***	0.0072***
	(0.00122)	(0.00122)	(0.00123)	(0.00123)	(0.00123)
Return	-0.0003	-0.0005	0.0013**	0.0011*	0.0014**
	(0.00060)	(0.00061)	(0.00060)	(0.00061)	(0.00060)
Debt/Equity Ratio	0.0182***	0.0178***			
	(0.00101)	(0.00101)			
Cash Surplus	-0.1087***	-0.1180***	-0.0944***	-0.1106***	-0.0893***
	(0.00536)	(0.00541)	(0.00564)	(0.00560)	(0.00556)
Log (CEO Age)	-0.0211***	-0.0205***	-0.0220***	-0.0214***	-0.0209***
	(0.00444)	(0.00453)	(0.00454)	(0.00460)	(0.00442)
Log (CEO Tenure)	-0.0026***	-0.0025***	-0.0019***	-0.0023***	-0.0020***
	(0.00063)	(0.00064)	(0.00064)	(0.00065)	(0.00063)
CEOGD	-0.0025	-0.0019	-0.0038	-0.0024	-0.0039
	(0.00382)	(0.00383)	(0.00405)	(0.00412)	(0.00390)
Constant	0.3324***	0.3352***	0.3386***	0.3489***	0.3249***
	(0.02172)	(0.02208)	(0.02103)	(0.02177)	(0.02071)
Channeling effect	0.00054***	0.00006	0.00147***	0.00007*	0.00191***
p-value	[0.000]	[0.123]	[0.000]	[0.0990]	[0.000]
% of total effect	11.44%	1.15%	24.22%	1.15%	31.47%

channeled					
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Adj. R-sq	0.443	0.434	0.433	0.416	0.448
N	29748	29748	29748	29748	29748

Panel B: Dependent variable - Volatility (ROA) and Volatility (ROE)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Volatility (ROA)					Volatility (ROE)				
SC(Total)	0.00148***	0.00176***	0.00053**	0.0008***	0.00026	0.02052***	0.02361***	0.00343	0.02154***	0.00047
	(0.00022)	(0.00023)	(0.00024)	(0.00023)	(0.00022)	(0.00342)	(0.00344)	(0.00304)	(0.00352)	(0.00296)
R&D/Sales	0.0050***				0.0051***	0.0576***				0.0521***
	(0.00052)				(0.00053)	(0.00833)				(0.00642)
Diversification		-0.0028***			-0.0030***		-0.0463***			-0.0555***
		(0.00065)			(0.00063)		(0.01239)			(0.01207)
Leverage			0.0069***		0.0134***			0.6237***		0.7074***
			(0.00154)		(0.00156)			(0.02876)		(0.03328)
Working Capital				0.0101***	0.0140***				-0.2193***	0.1800***
				(0.00172)	(0.00175)				(0.02666)	(0.02707)
Log (Age)	-0.0016***	-0.0018***	-0.0036***	-0.0031***	-0.0023***	-0.0125**	-0.0133**	-0.0388***	-0.0291***	-0.0224***
	(0.00032)	(0.00035)	(0.00036)	(0.00035)	(0.00034)	(0.00597)	(0.00659)	(0.00574)	(0.00633)	(0.00591)
Log (Size)	-0.0041***	-0.0044***	-0.0032***	-0.0029***	-0.0027***	-0.0169***	-0.0194***	-0.0230***	-0.0221***	-0.0163***
	(0.00016)	(0.00018)	(0.00016)	(0.00016)	(0.00015)	(0.00244)	(0.00264)	(0.00231)	(0.00278)	(0.00234)
MB Ratio	0.0004***	0.0005***	0.0008***	0.0008***	0.0007***	0.0075***	0.0083***	0.0088***	0.0089***	0.0070***
	(0.00006)	(0.00007)	(0.00008)	(0.00008)	(0.00007)	(0.00185)	(0.00185)	(0.00173)	(0.00194)	(0.00172)

Sales Growth	0.0029***	0.0033***	0.0040***	0.0039***	0.0035***	0.0249***	0.0297***	0.0312***	0.0290***	0.0259***
	(0.00048)	(0.00050)	(0.00052)	(0.00053)	(0.00049)	(0.00856)	(0.00870)	(0.00805)	(0.00878)	(0.00778)
Return	-0.0012***	-0.0014***	-0.0004*	-0.0006***	-0.0004**	-0.0139***	-0.0153***	-0.0126***	-0.0151***	-0.0130***
	(0.00019)	(0.00019)	(0.00020)	(0.00020)	(0.00019)	(0.00387)	(0.00394)	(0.00336)	(0.00372)	(0.00324)
Debt/Equity Ratio	-0.0008***	-0.0010***				0.0664***	0.0641***			
	(0.00018)	(0.00019)				(0.00725)	(0.00724)			
Cash Surplus	-0.0177***	-0.0224***	-0.0165***	-0.0214***	-0.0134***	-0.2246***	-0.2808***	-0.0739*	-0.2704***	-0.0483
	(0.00251)	(0.00260)	(0.00286)	(0.00286)	(0.00266)	(0.04572)	(0.04575)	(0.04065)	(0.04492)	(0.04023)
Log(CEO Age)	0.0011	0.0013	-0.0002	-0.0000	0.0002	0.0072	0.0106	0.0024	0.0103	0.0082
	(0.00148)	(0.00154)	(0.00161)	(0.00160)	(0.00152)	(0.02324)	(0.02350)	(0.02116)	(0.02379)	(0.02049)
Log(CEO Tenure)	-0.0009***	-0.0008***	-0.0006***	-0.0008***	-0.0007***	-0.0126***	-0.0122***	-0.0049	-0.0097***	-0.0058*
	(0.00020)	(0.00021)	(0.00021)	(0.00021)	(0.00020)	(0.00336)	(0.00342)	(0.00313)	(0.00347)	(0.00301)
CEOGD	-0.0026*	-0.0023	-0.0034**	-0.0028*	-0.0034**	-0.0095	-0.0056	-0.0286	-0.0121	-0.0272
	(0.00146)	(0.00157)	(0.00168)	(0.00162)	(0.00151)	(0.02667)	(0.02742)	(0.02478)	(0.02827)	(0.02324)
Constant	0.0430***	0.0446***	0.0470***	0.0425***	0.0367***	0.0093	0.0267	-0.0343	0.1527	-0.1618*
	(0.00849)	(0.00858)	(0.00800)	(0.00812)	(0.00753)	(0.09746)	(0.09871)	(0.09498)	(0.09884)	(0.09414)
Channeling effect	0.00030***	0.00002	0.00027***	0.00000***	0.00054***	0.00342***	0.00033	0.01975***	0.00164***	0.0227***

p-value	[0.000]	[0.117]	[0.000]	[0.009]	[0.000]	[0.000]	[0.126]	[0.000]	[0.008]	[0.000]
% of total effect channeled	20.27%	1.14%	33.75%	0.00%	67.5%	16.67%	1.39%	85.20%	7.08%	97.97%
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R-sq	0.347	0.310	0.265	0.267	0.326	0.123	0.105	0.239	0.092	0.269
N	29748	29748	29748	29748	29748	29748	29748	29748	29748	29748

Table 4: CEO Social Capital, Corporate Risk-taking – Second-Order Effects

This table reports estimates of coefficients from an OLS regression. The dependent variable is *Volatility (Return)* which is estimated as the volatility of monthly stock returns constructed over the window *t*+1 to *t*+3. *SC(Total)* is the natural logarithm of 1 plus number of individuals with whom the CEO is connected via educational, employment, or other social links in BoardEx universe. Educational links are estimated as the number of individuals with whom the CEO attended the same university, graduated within 2 years of each other, and earned a similar type of degree. Employment links are constructed as the number of individuals with whom the CEO shares a common employment history. Other social links are defined as the number of individuals with whom the CEO is connected through social clubs or other non-profit organizations. In Columns (1)-(2), (3)-(4), (5)-(6), (7)-(8), and (9)-(10) the sample is partitioned based on median values of *CEO Pay Slice*, *Free Cash Flow*, *External Financing*, *Information Asymmetry* and *Investment Efficiency*, respectively. *CEO Pay Slice* is the fraction of the aggregate compensation of the top-five executive team captured by the CEO. *Free Cash Flow* is estimated following Lehn and Poulsen (1989). *External Financing* is the annual change in book value of equity plus the change in deferred taxes minus the change in retained earnings, all scaled by lagged asset. *Information Asymmetry* is the absolute value of analysts' earnings forecast error for the fiscal year, estimated as the absolute value of actual earnings minus the earnings forecast scaled by the stock price. *Investment Efficiency* is estimated as the residual from a simple investment model (Biddle et al., 2009) that predicts the level of investment based on growth opportunities (measured by sales growth). Positive residual, as reflected in the error terms of the investment model, represent overinvestment and negative residual – underinvestment. Control variables are defined in the Table 1 header. Robust standard errors, adjusted for heteroskedasticity and clustered at the firm level, are reported in brackets below the coefficients. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	CEO Pay Slice		Free Cash Flow		External Financing		Information Asymmetry		Investment Efficiency	
	High	Low	High	Low	High	Low	High	Low	Overinvest	Underinvest
SC(Total)	0.0027**	0.0059***	0.0023***	0.0057***	0.0061***	0.0029***	0.0042***	0.0029***	0.0038***	0.0058***
	(0.00108)	(0.00110)	(0.00080)	(0.00098)	(0.00084)	(0.00084)	(0.00085)	(0.00074)	(0.00079)	(0.00089)
Log (Age)	-0.0101***	-0.0116***	-0.0091***	-0.0199***	-0.0179***	-0.0125***	-0.0116***	-0.0140***	-0.0144***	-0.0167***
	(0.00173)	(0.00171)	(0.00125)	(0.00177)	(0.00138)	(0.00132)	(0.00144)	(0.00128)	(0.00125)	(0.00149)
Log (Size)	-0.0107***	-0.0133***	-0.0097***	-0.0130***	-0.0154***	-0.0125***	-0.0135***	-0.0107***	-0.0128***	-0.0142***
	(0.00089)	(0.00080)	(0.00069)	(0.00091)	(0.00061)	(0.00059)	(0.00066)	(0.00059)	(0.00059)	(0.00063)

MB Ratio	0.0002	0.0003	0.0001	0.0001	0.0003	-0.0003	0.0003	0.0004**	0.0006***	0.0002
	(0.00023)	(0.00024)	(0.00019)	(0.00020)	(0.00019)	(0.00020)	(0.00022)	(0.00019)	(0.00020)	(0.00018)
Sales Growth	0.0057*	0.0119***	0.0135***	0.0023	0.0050***	-0.0025	0.0023	0.0195***	0.0119***	0.0024
	(0.00337)	(0.00342)	(0.00223)	(0.00154)	(0.00138)	(0.00276)	(0.00177)	(0.00242)	(0.00171)	(0.00174)
Return	0.0027*	-0.0007	0.0015	-0.0008	-0.0004	-0.0045***	-0.0017*	0.0022*	0.0004	-0.0008
	(0.00150)	(0.00148)	(0.00095)	(0.00083)	(0.00075)	(0.00120)	(0.00090)	(0.00125)	(0.00090)	(0.00082)
Debt/Equity Ratio	0.0226***	0.0214***	0.0192***	0.0174***	0.0193***	0.0186***	0.0172***	0.0177***	0.0178***	0.0185***
	(0.00209)	(0.00203)	(0.00146)	(0.00128)	(0.00158)	(0.00105)	(0.00094)	(0.00220)	(0.00132)	(0.00132)
Cash Surplus	-0.0967***	-0.1087***	-0.0698***	-0.1105***	-0.1061***	-0.1138***	-0.0839***	-0.0812***	-0.1448***	-0.1019***
	(0.01091)	(0.01112)	(0.00887)	(0.00673)	(0.00632)	(0.00804)	(0.00742)	(0.00788)	(0.00832)	(0.00634)
Log (CEO Age)	-0.0209***	-0.0215***	-0.0139***	-0.0146**	-0.0177***	-0.0197***	-0.0253***	-0.0123**	-0.0174***	-0.0220***
	(0.00738)	(0.00714)	(0.00528)	(0.00675)	(0.00571)	(0.00579)	(0.00592)	(0.00489)	(0.00561)	(0.00590)
Log (CEO Tenure)	0.0001	-0.0000	-0.0005	-0.0040***	-0.0032***	-0.0017**	-0.0019**	-0.0002	-0.0015*	-0.0034***
	(0.00100)	(0.00095)	(0.00067)	(0.00102)	(0.00085)	(0.00080)	(0.00086)	(0.00068)	(0.00079)	(0.00085)
CEOGD	-0.0108	-0.0014	-0.0036	0.0016	-0.0041	0.0011	0.0021	-0.0090**	-0.0048	-0.0000
	(0.00707)	(0.00575)	(0.00509)	(0.00540)	(0.00442)	(0.00481)	(0.00510)	(0.00425)	(0.00590)	(0.00447)
Constant	0.3385***	0.3323***	0.2767***	0.3009***	0.3458***	0.3127***	0.3796***	0.2788***	0.3058***	0.3485***
	(0.03166)	(0.02984)	(0.02377)	(0.03125)	(0.02646)	(0.02879)	(0.02592)	(0.01989)	(0.02599)	(0.02701)
Difference in coef.	0.0032**		0.0034***		-0.0032***		-0.0013		0.002**	

est. of SC(Total)										
p-value	[0.0189]		[0.0035]		[0.0035]		[0.1243]		[0.0465]	
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R-sq	0.442	0.437	0.425	0.335	0.449	0.418	0.408	0.438	0.457	0.418
N	6163	6171	13673	13681	13885	13892	10430	10438	14280	15453

Table 5: CEO Social Capital, Risk-taking and Corporate Risk-taking – Alternative Variable Measurement and Model Specification

This table reports estimates of coefficients from an OLS regression. In Panel A, the dependent variables are: in Columns (1)-(3) is *Volatility (Return)* which is estimated as the volatility of monthly stock returns constructed over the window *t*+1 to *t*+5, in Columns (4)-(6) *Volatility (ROA)* - estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+5, and in Columns (7)-(9) *Volatility (ROE)* - estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+5. Return on asset is the ratio of operating income to total assets and return on equity is the ratio of operating income to total stockholders' equity. In Panel B, the dependent variables are: in Columns (1)-(2) *R&D/Sales* - the ratio of research and development expenses to total sales, estimated as the average over the window *t*+1 to *t*+5, in Columns (3)-(4) *Diversification* - the measure of corporate diversification estimated following the methodology of Jacquemin and Berry (1979), constructed as the average over the window *t*+1 to *t*+5, in Columns (5)-(6) *Leverage* - the ratio of total debt to total assets, measured as the average over the window *t*+1 to *t*+5, in Columns (7)-(8) *Working Capital* - current assets minus current liabilities, scaled by total assets, measured as the average over the window *t*+1 to *t*+5. *SC(Total)* is the natural logarithm of 1 plus number of individuals with whom the CEO is connected via educational , employment, or other social links in BoardEx universe. Educational links are estimated as the number of individuals with whom the CEO attended the same university, graduated within 2 years of each other, and earned a similar type of degree. Employment links are constructed as the number of individuals with whom the CEO shares a common employment history. Other social links are defined as the number of individuals with whom the CEO is connected through social clubs or other non-profit organizations. *SC(Total)* $_{Excess}$ is estimated as the residuals from the regression of S*C(Total)* on the index of CEO human capital, defined as the sum of the following dummy variables: a dummy variable that takes the value of 1if a CEO has academic degree from "elite" college as defined by Engelberg et al. (2013) and zero otherwise, a dummy variable that takes the value of 1if a CEO has a Ph.D. and zero otherwise, a dummy variable that takes the value of 1if a CEO has legal experience and zero otherwise, a dummy variable that takes the value of 1if a CEO has finance experience and zero otherwise, a dummy variable that takes the value of 1if a CEO has political experience and zero otherwise, and a dummy variable that takes the value of 1if a CEO has military experience zero otherwise. *Log (Total Comp)* is the natural logarithm of the sum of salary and bonus compensation. *Delta* is the sensitivity of the value of the CEO's accumulated equity-based compensation to a one-percent change in the stock price. *Vega* is the sensitivity of the value of the CEO's accumulated equity-based compensation to a one-percent change in the volatility of stock prices. *Overconfidence* is estimated following Campbell et al. (2011). Other control variables are defined in the Table 1 header. Robust standard errors, adjusted for heteroskedasticity and clustered at the firm level, are reported in brackets below the coefficients. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: CEO Social Capital and Aggregate Corporate Risk-taking

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Corporate Risk-taking Measures								
	Volatility (Return) $_{t+1, t+5}$			Volatility (ROA) $_{t+1, t+5}$			Volatility (ROE) $_{t+1, t+5}$		
SC(Total) $_{Excess}$	0.0031**	0.0032***	0.0034***	0.0010***	0.0011***	0.0010***	0.0240***	0.0221***	0.0211***
	(0.00121)	(0.00119)	(0.00118)	(0.00037)	(0.00038)	(0.00038)	(0.00576)	(0.00551)	(0.00549)

Log(Age)	-0.0085***	-0.0084***	-0.0090***	-0.0015***	-0.0013***	-0.0014***	-0.0179**	-0.0155*	-0.0174**
	(0.00161)	(0.00156)	(0.00154)	(0.00045)	(0.00041)	(0.00042)	(0.00854)	(0.00854)	(0.00878)
Log(Size)	-0.0109***	-0.0105***	-0.0106***	-0.0033***	-0.0035***	-0.0035***	-0.0111**	-0.0085	-0.0088*
	(0.00088)	(0.00091)	(0.00092)	(0.00036)	(0.00036)	(0.00036)	(0.00527)	(0.00519)	(0.00526)
MB Ratio	0.0001	0.0001**	0.0001	0.0000	0.0000	0.0000	-0.0026**	-0.0027***	-0.0026**
	(0.00007)	(0.00006)	(0.00006)	(0.00002)	(0.00002)	(0.00002)	(0.00110)	(0.00102)	(0.00102)
Tangibility	0.0137***	0.0028	0.0053	0.0091***	0.0041	0.0047*	0.0641**	0.0442	0.0534
	(0.00502)	(0.00748)	(0.00731)	(0.00238)	(0.00249)	(0.00253)	(0.02877)	(0.03897)	(0.03955)
Sales Growth	0.0172***	0.0178***	0.0091***	0.0048***	0.0040***	0.0039***	-0.0001	0.0011	0.0022
	(0.00280)	(0.00284)	(0.00286)	(0.00107)	(0.00112)	(0.00120)	(0.01295)	(0.01292)	(0.01371)
Return	0.0023**	0.0015	0.0023**	0.0006**	0.0005*	0.0006**	0.0053	0.0065	0.0054
	(0.00093)	(0.00094)	(0.00092)	(0.00027)	(0.00027)	(0.00030)	(0.00416)	(0.00407)	(0.00440)
Debt/Equity Ratio	0.0105***	0.0095***	0.0103***	-0.0003**	-0.0002	-0.0001	0.0319***	0.0337***	0.0350***
	(0.00173)	(0.00157)	(0.00167)	(0.00015)	(0.00013)	(0.00014)	(0.00792)	(0.00769)	(0.00779)
Free Cash Flow	-0.1253***	-0.1310***	-0.1276***	-0.0130	-0.0179**	-0.0182**	-0.0873	-0.0987	-0.0995
	(0.01030)	(0.01053)	(0.01040)	(0.00825)	(0.00796)	(0.00800)	(0.08974)	(0.08725)	(0.08726)
Log(CEO Age)	-0.0166*	-0.0170**	-0.0171**	0.0019	0.0004	0.0003	-0.0126	-0.0070	-0.0092
	(0.00847)	(0.00807)	(0.00793)	(0.00243)	(0.00218)	(0.00219)	(0.03738)	(0.03824)	(0.03831)
Log(CEO Tenure)	0.0002	0.0001	0.0003	-0.0008***	-0.0006**	-0.0006**	-0.0140***	-0.0150***	-0.0151***

	(0.00100)	(0.00093)	(0.00092)	(0.00030)	(0.00028)	(0.00028)	(0.00524)	(0.00530)	(0.00534)
CEOGD	-0.0228**	-0.0225***	-0.0208**	-0.0040*	-0.0019	-0.0016	0.0211	0.0576***	0.0613***
	(0.00925)	(0.00855)	(0.00847)	(0.00228)	(0.00202)	(0.00205)	(0.01749)	(0.02055)	(0.02080)
Log(Total Comp)	-0.0020	-0.0005	-0.0017	0.0015***	0.0012***	0.0014***	0.0156**	0.0150**	0.0156**
	(0.00125)	(0.00125)	(0.00133)	(0.00044)	(0.00041)	(0.00044)	(0.00747)	(0.00693)	(0.00755)
Delta	0.0001	0.0001	0.0000	0.0000***	0.0000*	0.0000*	0.0006	0.0005	0.0006
	(0.00007)	(0.00008)	(0.00006)	(0.00002)	(0.00002)	(0.00002)	(0.00041)	(0.00045)	(0.00046)
Vega	-0.0023	-0.0054**	-0.0031	0.0017*	0.0020**	0.0021**	-0.0243**	-0.0350**	-0.0353**
	(0.00270)	(0.00270)	(0.00253)	(0.00097)	(0.00086)	(0.00086)	(0.01067)	(0.01362)	(0.01405)
Overconfidence	0.0006	0.0009	0.0009	0.0006	0.0009	0.0007	0.0040	0.0083	0.0060
	(0.00222)	(0.00210)	(0.00212)	(0.00072)	(0.00069)	(0.00069)	(0.01158)	(0.01133)	(0.01135)
Constant	0.3100***	0.3099***	0.3420***	0.0232**	0.0448***	0.0441***	0.1466	0.0059	-0.0130
	(0.03390)	(0.03634)	(0.03479)	(0.00982)	(0.01112)	(0.01119)	(0.15065)	(0.15858)	(0.16007)
Industry FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Year FE	No	No	Yes	No	No	Yes	No	No	Yes
Adj. R-sq	0.228	0.298	0.364	0.125	0.235	0.238	0.046	0.097	0.099

N	7943	7943	7943	7943	7943	7943	7943	7943	7943

Panel B: CEO Social Capital and Corporate Policies

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Corporate Policies							
	$R\&D_{t+1,\ t+5}$		$Diversification_{t+1,\ t+5}$		$Leverage_{t+1,\ t+5}$		$Working\ Capital_{t+1,\ t+5}$	
$SC(Total)_{Excess}$	0.0207*	0.0189*	-0.0189**	-0.0144*	0.0367***	0.0365***	-0.0178***	-0.0178***
	(0.01235)	(0.01072)	(0.00858)	(0.00870)	(0.00501)	(0.00507)	(0.00378)	(0.00385)
Log(Age)	-0.0235	-0.0254	0.1493***	0.1561***	0.0242***	0.0240***	-0.0136***	-0.0139***
	(0.03119)	(0.02906)	(0.01327)	(0.01350)	(0.00677)	(0.00689)	(0.00526)	(0.00535)
Log(Size)	-0.0617*	-0.0634	0.0549***	0.0564***	0.0053	0.0046	-0.0217***	-0.0216***
	(0.03737)	(0.03904)	(0.00795)	(0.00793)	(0.00388)	(0.00399)	(0.00283)	(0.00291)
MB Ratio	0.0010	0.0011	-0.0007	-0.0008*	-0.0008	-0.0008	0.0004*	0.0004*
	(0.00127)	(0.00137)	(0.00045)	(0.00049)	(0.00052)	(0.00052)	(0.00023)	(0.00023)
Tangibility	-0.2952***	-0.2863***	-0.0172	-0.0469	0.1410***	0.1389***	-0.3031***	-0.3012***
	(0.10050)	(0.09660)	(0.05567)	(0.05632)	(0.03263)	(0.03284)	(0.02511)	(0.02542)
Sales Growth	0.1402	0.1503	0.0139	0.0075	-0.0089	-0.0064	-0.0027	-0.0053
	(0.12470)	(0.13624)	(0.01031)	(0.01054)	(0.00922)	(0.00967)	(0.00663)	(0.00713)

Return	0.0193	0.0198	0.0072*	0.0066	-0.0042	-0.0061**	0.0114***	0.0133***
	(0.01970)	(0.02145)	(0.00385)	(0.00425)	(0.00273)	(0.00299)	(0.00194)	(0.00216)
Debt/Equity Ratio	-0.0039	-0.0037	-0.0162***	-0.0182***				
	(0.00302)	(0.00320)	(0.00434)	(0.00450)				
Free Cash Flow	-1.5017**	-1.5165**	-0.1734***	-0.1576***	-0.4102***	-0.4121***	0.0588	0.0593
	(0.75501)	(0.76933)	(0.05184)	(0.05117)	(0.07884)	(0.07936)	(0.04116)	(0.04126)
Log(CEO Age)	0.1269	0.1246	0.1010*	0.1113*	-0.0190	-0.0190	0.0434	0.0435
	(0.16248)	(0.15794)	(0.05963)	(0.06007)	(0.03393)	(0.03412)	(0.02863)	(0.02878)
Log(CEO Tenure)	-0.0048	-0.0057	-0.0008	0.0006	-0.0106***	-0.0110***	0.0123***	0.0124***
	(0.01401)	(0.01487)	(0.00712)	(0.00712)	(0.00395)	(0.00397)	(0.00333)	(0.00334)
CEOGD	0.0676*	0.0710*	0.0802**	0.0683*	0.0106	0.0105	-0.0098	-0.0091
	(0.03616)	(0.03948)	(0.03559)	(0.03540)	(0.03239)	(0.03240)	(0.02223)	(0.02228)
Log(Total Comp)	0.0238	0.0302	0.0276**	0.0227*	0.0583***	0.0619***	-0.0356***	-0.0367***
	(0.02191)	(0.02848)	(0.01121)	(0.01190)	(0.00621)	(0.00687)	(0.00460)	(0.00496)
Delta	0.0009	0.0010	-0.0001	-0.0003	0.0001	0.0001	0.0007**	0.0007**
	(0.00066)	(0.00076)	(0.00080)	(0.00076)	(0.00030)	(0.00030)	(0.00030)	(0.00031)
Vega	0.0637*	0.0660*	-0.0129	-0.0176	-0.0464***	-0.0455***	0.0029	0.0036
	(0.03391)	(0.03624)	(0.04150)	(0.04062)	(0.01705)	(0.01721)	(0.00986)	(0.00997)
Overconfidence	0.0722	0.0696	-0.0497***	-0.0413**	-0.0211**	-0.0201**	0.0161*	0.0153*

	(0.05774)	(0.05404)	(0.01783)	(0.01807)	(0.00995)	(0.01010)	(0.00837)	(0.00848)
Constant	-0.1057	-0.1525	-1.0197***	-0.9589***	0.0701	0.0706	0.5791***	0.5746***
	(0.64347)	(0.69285)	(0.24054)	(0.24149)	(0.14035)	(0.14093)	(0.11695)	(0.11661)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes
Adj. R-sq	0.114	0.114	0.335	0.342	0.349	0.351	0.515	0.515
N	7943	7943	7943	7943	7943	7943	7943	7943

Table 6: CEO Social Capital, Corporate Risk-taking, and Corporate Policies - Disentangling the Channels - Alternative Variable Measurement and Model Specification

This table reports estimates of coefficients from an OLS regression. The dependent variable in Panel A is Columns *Volatility (Return)* which is estimated as the volatility of monthly stock returns constructed over the window *t*+1 to *t*+5. In Panel B the dependent variables are in Columns (1)-(5) *Volatility (ROA)* which is estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+5, and in Columns (6)-(10) *Volatility (ROE)* which is estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+5. Return on asset is the ratio of operating income to total assets and return on equity is the ratio of operating income to total stockholders' equity. *SC(Total)* is the natural logarithm of 1 plus number of individuals with whom the CEO is connected via educational, employment, or other social links in BoardEx universe. Educational links are estimated as the number of individuals with whom the CEO attended the same university, graduated within 2 years of each other, and earned a similar type of degree. Employment links are constructed as the number of individuals with whom the CEO shares a common employment history. Other social links are defined as the number of individuals with whom the CEO is connected through social clubs or other non-profit organizations. $SC(Total)_{Excess}$ is estimated as the residuals from the regression of S*C(Total)* on the index of CEO human capital, defined as the sum of the following dummy variables: a dummy variable that takes the value of 1if a CEO has academic degree from "elite" college as defined by Engelberg et al. (2013) and zero otherwise, a dummy variable that takes the value of 1if a CEO has a Ph.D. and zero otherwise, a dummy variable that takes the value of 1if a CEO has legal experience and zero otherwise, a dummy variable that takes the value of 1if a CEO has finance experience and zero otherwise, a dummy variable that takes the value of 1if a CEO has political experience and zero otherwise, and a dummy variable that takes the value of 1if a CEO has military experience zero otherwise. *R&D/Sales* is the ratio of research and development expenses to total sales, estimated as the average over the window *t*+1 to *t*+5. *Diversification* is the measure of corporate diversification estimated following the methodology of Jacquemin and Berry (1979), constructed as the average over the window *t*+1 to *t*+5. *Leverage* is the ratio of total debt to total assets, measured as the average over the window *t*+1 to *t*+5. *Working Capital* is current assets minus current liabilities, scaled by total assets, measured as the average over the window *t*+1 to *t*+5. *Log(Total Comp)* is the natural logarithm of the sum of salary and bonus compensation. *Delta* is the sensitivity of the value of the CEO's accumulated equity-based compensation to a one-percent change in the stock price. *Vega* is the sensitivity of the value of the CEO's accumulated equity-based compensation to a one-percent change in the volatility of stock prices. *Overconfidence* is estimated following Campbell et al. (2011). Other *c*ontrol variables are defined in the Table 1 header. Channeling effect is the reduction in the effects of CEO social capital on return and earnings volatilities, estimated as the change in the coefficient estimates of social capital from the respective values reported in Table 5, Panel A. Robust standard errors, adjusted for heteroskedasticity and clustered at the firm level, are reported in brackets below the coefficients. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: Dependent variable - Volatility (Return) $_{t+1, t+5}$

	(1)	(2)	(3)	(4)	(5)
$SC(Total)_{Excess}$	0.0033***	0.0033***	0.0031***	0.0049***	0.0031***
	(0.00117)	(0.00118)	(0.00110)	(0.00118)	(0.00110)
R&D Sales Ratio	0.0054***				0.0041*
	(0.00204)				(0.00240)
Diversification		-0.0064**			-0.0090***

		(0.00259)			(0.00258)
Leverage			0.0517***		0.0643***
			(0.00623)		(0.00720)
Working Capital				-0.0025	0.0320***
				(0.00807)	(0.00860)
Log(Age)	-0.0089***	-0.0080***	-0.0108***	-0.0096***	-0.0090***
	(0.00154)	(0.00158)	(0.00151)	(0.00155)	(0.00153)
Log(Size)	-0.0103***	-0.0102***	-0.0135***	-0.0133***	-0.0125***
	(0.00092)	(0.00091)	(0.00083)	(0.00088)	(0.00085)
MB Ratio	0.0000	0.0000	0.0002***	0.0002***	0.0002***
	(0.00006)	(0.00006)	(0.00007)	(0.00007)	(0.00007)
Tangibility	0.0069	0.0050	-0.0036	0.0028	0.0054
	(0.00731)	(0.00730)	(0.00737)	(0.00805)	(0.00781)
Sales Growth	0.0083***	0.0091***	0.0118***	0.0115***	0.0115***
	(0.00290)	(0.00286)	(0.00279)	(0.00291)	(0.00285)
Return	0.0022**	0.0024***	0.0044***	0.0041***	0.0039***
	(0.00091)	(0.00092)	(0.00091)	(0.00093)	(0.00090)
Debt/Equity Ratio	0.0103***	0.0101***			
	(0.00168)	(0.00167)			
Free Cash Flow	-0.1194***	-0.1287***	-0.0888***	-0.1100***	-0.0816***
	(0.01073)	(0.01037)	(0.01054)	(0.01030)	(0.01062)
Log(CEO Age)	-0.0178**	-0.0164**	-0.0153**	-0.0161**	-0.0158**
	(0.00794)	(0.00794)	(0.00770)	(0.00801)	(0.00771)
Log(CEO Tenure)	0.0003	0.0003	0.0007	0.0002	0.0004
	(0.00092)	(0.00092)	(0.00089)	(0.00092)	(0.00088)
CEOGD	-0.0212**	-0.0204**	-0.0198**	-0.0193**	-0.0192**

	(0.00846)	(0.00844)	(0.00855)	(0.00909)	(0.00845)
Log(Total Comp)	-0.0019	-0.0016	-0.0030**	0.0001	-0.0023*
	(0.00133)	(0.00131)	(0.00135)	(0.00129)	(0.00131)
Delta	0.0000	0.0000	0.0001	0.0001	0.0001
	(0.00006)	(0.00006)	(0.00008)	(0.00007)	(0.00007)
Vega	-0.0035	-0.0033	0.0037	0.0013	0.0042
	(0.00252)	(0.00253)	(0.00274)	(0.00274)	(0.00271)
Overconfidence	0.0005	0.0006	0.0049**	0.0039*	0.0039*
	(0.00211)	(0.00211)	(0.00215)	(0.00216)	(0.00213)
Constant	0.3429***	0.3359***	0.3404***	0.3455***	0.3130***
	(0.03491)	(0.03486)	(0.03198)	(0.03416)	(0.03270)
Channeling effect	0.0001**	0.0001	0.0019***	0.00004	0.0019***
p-value	[0.028]	[0.215]	[0.000]	[0.61]	[0.000]
% of total effect channeled	2.94%	2.94%	38.25%	0.89%	37.50%
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Adj. R-sq	0.367	0.365	0.392	0.364	0.403
N	7943	7943	7943	7943	7943

Panel B: Dependent variable - Volatility (ROA) $_{t+1, t+5}$ and Volatility (ROE) $_{t+1, t+5}$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Volatility (ROA) $_{t+1, t+5}$					Volatility (ROE) $_{t+1, t+5}$				
$SC(Total)_{Excess}$	0.0009**	0.0098***	0.0002	0.0005**	0.0002	0.0202***	0.0205***	0.0021	0.0214***	0.0026
	(0.00038)	(0.00038)	(0.00034)	(0.00034)	(0.00033)	(0.00554)	(0.00549)	(0.00524)	(0.00571)	(0.00515)
R&D Sales Ratio	0.0045***				0.0041***	0.0483***				0.0331**
	(0.00055)				(0.00055)	(0.01772)				(0.01441)
Diversification		-0.0036***			-0.0036***		-0.0428***			-0.0542***
		(0.00073)			(0.00073)		(0.01494)			(0.01425)
Leverage			0.0065**		0.0136***			0.5875***		0.6922***
			(0.00325)		(0.00366)			(0.04789)		(0.05709)
Working Capital				0.0141***	0.0189***				-0.1177***	0.2742***
				(0.00244)	(0.00309)				(0.04017)	(0.04501)
Log(Age)	-0.0013***	-0.0009**	-0.0025***	-0.0021***	-0.0016***	-0.0162*	-0.0107	-0.0328***	-0.0203**	-0.0209***
	(0.00041)	(0.00042)	(0.00047)	(0.00044)	(0.00044)	(0.00872)	(0.00905)	(0.00788)	(0.00881)	(0.00800)
Log(Size)	-0.0032***	-0.0033***	-0.0019***	-0.0016***	-0.0014***	-0.0057	-0.0064	-0.0110***	-0.0109**	-0.0032
	(0.0003	(0.0003	(0.0002	(0.0002	(0.0002	(0.0053	(0.0053	(0.0041	(0.0049	(0.0042

	6)	6)	6)	6)	5)	3)	8)	7)	6)	0)
MB Ratio	0.0000	0.0000	0.0001**	0.0001*	0.0001**	-0.0027***	-0.0027***	-0.0022**	-0.0026**	-0.0023***
	(0.00002)	(0.00002)	(0.00003)	(0.00003)	(0.00002)	(0.00102)	(0.00102)	(0.00086)	(0.00103)	(0.00085)
Tangibility	0.0060**	0.0045*	0.0016	0.0067***	0.0076***	0.0672*	0.0513	-0.0189	0.0273	0.0590
	(0.00250)	(0.00252)	(0.00247)	(0.00248)	(0.00245)	(0.03848)	(0.03951)	(0.03855)	(0.04345)	(0.03871)
Sales Growth	0.0032**	0.0039***	0.0042***	0.0042***	0.0037***	-0.0050	0.0026	0.0050	0.0007	0.0023
	(0.00127)	(0.00120)	(0.00119)	(0.00118)	(0.00120)	(0.01447)	(0.01370)	(0.01282)	(0.01373)	(0.01279)
Return	0.0005**	0.0006**	0.0014***	0.0011***	0.0010***	0.0044	0.0056	0.0075*	0.0055	0.0033
	(0.00026)	(0.00030)	(0.00032)	(0.00032)	(0.00027)	(0.00429)	(0.00440)	(0.00421)	(0.00444)	(0.00407)
Debt/Equity Ratio	-0.0001	-0.0002				0.0352***	0.0342***			
	(0.00013)	(0.00014)				(0.00782)	(0.00774)			
Free Cash Flow	-0.0114	-0.0188**	-0.0099	-0.0134	-0.0029	-0.0262	-0.1062	0.1096	-0.1255	0.1723*
	(0.00795)	(0.00798)	(0.00826)	(0.00905)	(0.00774)	(0.08849)	(0.08683)	(0.09023)	(0.08576)	(0.09060)
Log(CEO Age)	-0.0003	0.0007	-0.0004	-0.0011	-0.0011	-0.0152	-0.0044	0.0088	0.0027	0.0018
	(0.00210)	(0.00219)	(0.00228)	(0.00225)	(0.00210)	(0.03808)	(0.03879)	(0.03455)	(0.03893)	(0.03452)
Log(CEO Tenure)	-0.0006**	-0.0006**	-0.0004	-0.0006**	-0.0005**	-0.0148***	-0.0151***	-0.0081*	-0.0131**	-0.0105**
	(0.0002	(0.0002	(0.0002	(0.0002	(0.0002	(0.0053	(0.0053	(0.0049	(0.0052	(0.0046

	7)	8)	8)	7)	6)	6)	6)	2)	5)	2)
CEOGD	-0.0019	-0.0014	-0.0017	-0.0016	-0.0017	0.0578***	0.0642***	0.0514**	0.0565***	0.0546**
	(0.00205)	(0.00206)	(0.00218)	(0.00209)	(0.00214)	(0.02080)	(0.02089)	(0.02148)	(0.02172)	(0.02122)
Log(Total Comp)	0.0013***	0.0015***	-0.0004	0.0005	0.0000	0.0142*	0.0166**	-0.0177***	0.0144**	-0.0117*
	(0.00044)	(0.00045)	(0.00045)	(0.00039)	(0.00042)	(0.00748)	(0.00755)	(0.00675)	(0.00721)	(0.00634)
Delta	0.0000*	0.0000*	0.0000	0.0000	-0.0000	0.0005	0.0006	0.0005	0.0007	0.0003
	(0.00002)	(0.00002)	(0.00002)	(0.00001)	(0.00001)	(0.00045)	(0.00046)	(0.00040)	(0.00044)	(0.00039)
Vega	0.0018**	0.0021**	0.0009	0.0006	0.0011	-0.0384***	-0.0360**	-0.0158	-0.0421***	-0.0120
	(0.00083)	(0.00087)	(0.00077)	(0.00074)	(0.00078)	(0.01398)	(0.01422)	(0.01263)	(0.01449)	(0.01297)
Overconfidence	0.0004	0.0005	0.0014*	0.0010	0.0007	0.0027	0.0042	0.0150	0.0050	0.0079
	(0.00065)	(0.00068)	(0.00072)	(0.00071)	(0.00064)	(0.01118)	(0.01129)	(0.01027)	(0.01136)	(0.00971)
Constant	0.0447***	0.0407***	0.0452***	0.0376***	0.0309***	-0.0056	-0.0540	-0.0807	0.0284	-0.2936**
	(0.01089)	(0.01123)	(0.01120)	(0.01105)	(0.01055)	(0.15927)	(0.16425)	(0.14779)	(0.15891)	(0.14515)
Channeling effect	0.0001**	0.00002	0.0003**	0.000	0.0003*	0.0009**	0.0006	0.021***	0.0021	0.0209***
p-value	[0.045]	[0.127]	[0.028]	[0.9]	[0.068]	[0.016]	[0.196]	[0.000]	[0.100]	[0.000]
% of total effect	11.11%	2.0%	55.85%	0.00%	55.85%	4.46%	2.84%	91.08%	9.05%	88.95%

channeled										
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R-sq	0.269	0.243	0.205	0.216	0.274	0.112	0.102	0.254	0.083	0.288
N	7943	7943	7943	7943	7943	7943	7943	7943	7943	7943

Table 7: CEO Social Capital, Corporate Risk-taking – Second-Order Effects: Alternative Variable Measurement and Model Specification

This table reports estimates of coefficients from an OLS regression. The dependent variable is *Volatility (Return)* which is estimated as the volatility of monthly stock returns constructed over the window *t*+1 to *t*+5. *SC(Total)* is the natural logarithm of 1 plus number of individuals with whom the CEO is connected via educational, employment, or other social links in BoardEx universe. Educational links are estimated as the number of individuals with whom the CEO attended the same university, graduated within 2 years of each other, and earned a similar type of degree. Employment links are constructed as the number of individuals with whom the CEO shares a common employment history. Other social links are defined as the number of individuals with whom the CEO is connected through social clubs or other non-profit organizations. $SC(Total)_{Excess}$ is estimated as the residuals from the regression of S*C(Total)* on the index of CEO human capital, defined as the sum of the following dummy variables: a dummy variable that takes the value of 1if a CEO has academic degree from "elite" college as defined by Engelberg et al. (2013) and zero otherwise, a dummy variable that takes the value of 1if a CEO has a Ph.D. and zero otherwise, a dummy variable that takes the value of 1if a CEO has legal experience and zero otherwise, a dummy variable that takes the value of 1if a CEO has finance experience and zero otherwise, a dummy variable that takes the value of 1if a CEO has political experience and zero otherwise, and a dummy variable that takes the value of 1if a CEO has military experience zero otherwise. In Panel A, Columns (1)-(2), (3)-(4), (5)-(6), and (7)-(8) the sample is partitioned based on median values of *GIM Index*, *BCF Index, Co-option,* and *Hostile Takeover Threat,* respectively. *GIM Index* is from Gompers et al. (2003), *BCF Index* is from Bebchuk et al. (2009), *Co-option* – from Coles et al. (2014), and *Hostile Takeover Threat* – from Cain et al. (2017). In Panel B, Columns (1)-(2), (3)-(4), (5)-(6), and (7)-(8) the sample is partitioned based on median values of *Free Cash Flow*, *External Financing*, *Information Asymmetry* and *Investment Efficiency,* respectively. *Free Cash Flow* is estimated following Lehn and Poulsen (1989). *External Financing* is the annual change in book value of equity plus the change in deferred taxes minus the change in retained earnings, all scaled by lagged asset. *Information Asymmetry* is the absolute value of analysts' earnings forecast error for the fiscal year, estimated as the absolute value of actual earnings minus the earnings forecast scaled by the stock price. *Investment Efficiency* is estimated as the residual from a simple investment model (Biddle et al., 2009) that predicts the level of investment based on growth opportunities (measured by sales growth). Positive residual, as reflected in the error terms of the investment model, represent overinvestment and negative residual – underinvestment. *Log (Total Comp)* is the natural logarithm of the sum of salary and bonus compensation. *Delta* is the sensitivity of the value of the CEO's accumulated equity-based compensation to a one-percent change in the stock price. *Vega* is the sensitivity of the value of the CEO's accumulated equity-based compensation to a one-percent change in the volatility of stock prices. *Overconfidence* is estimated following Campbell et al. (2011). Other control variables are defined in the Table 1 header. Robust standard errors, adjusted for heteroskedasticity and clustered at the firm level, are reported in brackets below the coefficients. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A: Managerial Entrenchment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	GIM Index		BCF Index		Co-option		Hostile Takeover Threat	
	High	Low	High	Low	High	Low	High	Low
$SC(Total)_{Excess}$	0.0015	0.0042*	-0.0035	0.0051***	0.0019	0.0046***	0.0052***	0.0009
	(0.00207)	(0.00228)	(0.00331)	(0.00176)	(0.00168)	(0.00135)	(0.00174)	(0.00156)
Log(Age)	-0.0047*	-0.0133***	-0.0053	-0.0091***	-0.0095***	-0.0089***	-0.0114***	-0.0105***
	(0.00274)	(0.00301)	(0.00350)	(0.00230)	(0.00225)	(0.00229)	(0.00391)	(0.00304)
Log(Size)	-0.0137***	-0.0103***	-0.0116***	-0.0117***	-0.0068***	-0.0119***	-0.0104***	-0.0114***
	(0.00175)	(0.00197)	(0.00242)	(0.00150)	(0.00136)	(0.00136)	(0.00119)	(0.00150)
MB Ratio	-0.0002	-0.0001	-0.0003	-0.0001	-0.0001	0.0000	-0.0001	0.0002*
	(0.00018)	(0.00019)	(0.00088)	(0.00015)	(0.00011)	(0.00008)	(0.00011)	(0.00011)

Tangibility	-0.0081	0.0131	-0.0274	0.0034	0.0041	0.0202**	-0.0027	0.0028
	(0.01316)	(0.01481)	(0.01778)	(0.01174)	(0.01032)	(0.01008)	(0.01014)	(0.01055)
Sales Growth	0.0003	0.0083*	0.0076	0.0070*	0.0061*	0.0056	-0.0030	0.0116***
	(0.00644)	(0.00466)	(0.00901)	(0.00402)	(0.00344)	(0.00448)	(0.00443)	(0.00375)
Return	-0.0041	-0.0081**	-0.0016	-0.0102***	-0.0016	-0.0003	0.0010	0.0025**
	(0.00386)	(0.00369)	(0.00540)	(0.00318)	(0.00212)	(0.00228)	(0.00153)	(0.00112)
Debt/Equity Ratio	0.0172***	0.0063***	0.0152***	0.0076***	0.0070***	0.0120***	0.0153***	0.0103***
	(0.00338)	(0.00207)	(0.00382)	(0.00232)	(0.00230)	(0.00222)	(0.00256)	(0.00171)
Free Cash Flow	-0.1648***	-0.1160***	-0.1363***	-0.1072***	-0.1458***	-0.1383***	-0.1088***	-0.1355***
	(0.02508)	(0.02069)	(0.03327)	(0.01723)	(0.01743)	(0.01454)	(0.01415)	(0.01464)
Log(CEO Age)	-0.0197	0.0021	-0.0269	0.0020	-0.0334**	0.0016	0.0005	-0.0300**
	(0.01365)	(0.01595)	(0.02196)	(0.01233)	(0.01302)	(0.00964)	(0.00958)	(0.01224)
Log(CEO Tenure)	0.0013	-0.0012	-0.0012	-0.0004	0.0020	-0.0005	-0.0007	0.0004
	(0.00182)	(0.00174)	(0.00252)	(0.00144)	(0.00126)	(0.00154)	(0.00116)	(0.00144)

CEOGD	-0.0611***	-0.0259	-0.1130***	-0.0274**	-0.0076	-0.0252**	-0.0414***	0.0014
	(0.01833)	(0.01700)	(0.02411)	(0.01352)	(0.00903)	(0.01120)	(0.01249)	(0.00849)
Log(Total Comp)	0.0018	-0.0004	0.0008	-0.0012	-0.0041**	0.0004	0.0010	-0.0037**
	(0.00273)	(0.00234)	(0.00293)	(0.00209)	(0.00158)	(0.00169)	(0.00186)	(0.00183)
Delta	0.0003***	-0.0000	0.0034	0.0001**	0.0001**	0.0002	0.0000	0.0002**
	(0.00008)	(0.00006)	(0.00247)	(0.00005)	(0.00004)	(0.00014)	(0.00006)	(0.00007)
Vega	-0.0041	-0.0021	0.0090	-0.0087**	-0.0070**	0.0013	-0.0070**	-0.0047
	(0.00523)	(0.00488)	(0.01028)	(0.00366)	(0.00273)	(0.00431)	(0.00290)	(0.00515)
Overconfidence	0.0074*	-0.0013	-0.0020	-0.0013	0.0050*	-0.0006	0.0042	0.0009
	(0.00446)	(0.00392)	(0.00614)	(0.00318)	(0.00299)	(0.00290)	(0.00345)	(0.00305)
Constant	0.3481***	0.2641***	0.5235***	0.2829***	0.3492***	0.2479***	0.3051***	0.3792***
	(0.05417)	(0.06450)	(0.08833)	(0.05132)	(0.05285)	(0.03844)	(0.04134)	(0.04826)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Adj. R-sq	0.438	0.319	0.380	0.389	0.340	0.371	0.392	0.378
N	1002	1188	616	1592	2991	3139	3664	3738

Panel B: Other Second-Order Effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Free cash flow		External Financing		Information Asymmetry		Investment Efficacy	
	High	Low	High	Low	High	Low	Overinvest	Underinvest
$SC(Total)_{Excess}$	0.0022	0.0039**	0.0044***	0.0015	0.0049***	0.0011	0.0020	0.0039***
	(0.00150)	(0.00168)	(0.00149)	(0.00147)	(0.00166)	(0.00102)	(0.00162)	(0.00121)
Log(Age)	-0.0027	-0.0124***	-0.0117***	-0.0072***	-0.0079***	-0.0086***	-0.0088***	-0.0085***
	(0.00184)	(0.00227)	(0.00172)	(0.00206)	(0.00208)	(0.00166)	(0.00254)	(0.00161)
Log(Size)	-0.0074***	-0.0081***	-0.0103***	-0.0105***	-0.0122***	-0.0087***	-0.0103***	-0.0107***
	(0.00137)	(0.00179)	(0.00113)	(0.00122)	(0.00129)	(0.00097)	(0.00143)	(0.00097)
MB Ratio	-0.0000	0.0001	0.0002*	-0.0000	0.0001	0.0000	0.0001	0.0000
	(0.00005)	(0.00013)	(0.00011)	(0.00007)	(0.00009)	(0.00007)	(0.00011)	(0.00009)
Tangibility	0.0069	0.0071	-0.0025	0.0098	0.0058	-0.0047	-0.0011	0.0090
	(0.00848)	(0.01071)	(0.00837)	(0.01057)	(0.00972)	(0.00733)	(0.00984)	(0.00796)

Sales Growth	0.0127***	0.0063	0.0100***	-0.0184***	-0.0016	0.0238***	0.0091**	0.0066*
	(0.00369)	(0.00414)	(0.00334)	(0.00529)	(0.00317)	(0.00435)	(0.00427)	(0.00366)
Return	0.0015	0.0005	0.0022**	-0.0026	0.0013	0.0020	0.0006	0.0031***
	(0.00155)	(0.00115)	(0.00098)	(0.00174)	(0.00137)	(0.00142)	(0.00172)	(0.00116)
Debt/Equity Ratio	0.0089***	0.0098***	0.0145***	0.0097***	0.0082***	0.0199***	0.0104***	0.0101***
	(0.00286)	(0.00143)	(0.00276)	(0.00170)	(0.00152)	(0.00380)	(0.00346)	(0.00131)
Free Cash Flow	-0.0857***	-0.1289***	-0.1218***	-0.1046***	-0.1020***	-0.0574***	-0.1114***	-0.1341***
	(0.01509)	(0.01304)	(0.01296)	(0.01490)	(0.01232)	(0.01406)	(0.01432)	(0.01207)
Log(CEO Age)	-0.0133	-0.0176	-0.0112	-0.0168	-0.0139	-0.0050	-0.0097	-0.0197**
	(0.00935)	(0.01104)	(0.00930)	(0.01097)	(0.00984)	(0.00767)	(0.01059)	(0.00838)
Log(CEO Tenure)	0.0017	-0.0007	0.0000	0.0003	-0.0002	0.0009	-0.0010	0.0009
	(0.00111)	(0.00134)	(0.00115)	(0.00122)	(0.00128)	(0.00093)	(0.00136)	(0.00103)
CEOGD	-0.0253**	-0.0159	-0.0267**	-0.0126	-0.0281***	-0.0336***	-0.0267**	-0.0192**
	(0.01278)	(0.01019)	(0.01259)	(0.00866)	(0.00920)	(0.01157)	(0.01254)	(0.00850)
Log(Total Current Comp)	-0.0012	-0.0010	-0.0027*	0.0006	-0.0023	-0.0009	-0.0003	-0.0019
	(0.0013	(0.0021	(0.0016	(0.0019	(0.0020	(0.0012	(0.0018	(0.0014

	7)	0)	2)	8)	7)	8)	7)	6)
Delta	0.0000	0.0004***	0.0000	-0.0000	0.0001	0.0001***	0.0001	0.0001
	(0.00005)	(0.00010)	(0.00004)	(0.00008)	(0.00007)	(0.00004)	(0.00012)	(0.00006)
Vega	-0.0055**	-0.0404***	-0.0047*	-0.0020	-0.0063	-0.0011	0.0018	-0.0050**
	(0.00250)	(0.01462)	(0.00278)	(0.00339)	(0.00697)	(0.00207)	(0.00566)	(0.00250)
Overconfidence	0.0008	0.0008	-0.0000	-0.0010	0.0012	0.0011	-0.0000	0.0007
	(0.00255)	(0.00318)	(0.00240)	(0.00301)	(0.00297)	(0.00200)	(0.00306)	(0.00227)
Constant	0.2616***	0.3565***	0.3762***	0.3462***	0.3673***	0.2435***	0.2927***	0.3693***
	(0.04124)	(0.04455)	(0.03882)	(0.04383)	(0.04109)	(0.03193)	(0.04375)	(0.03728)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R-sq	0.309	0.339	0.406	0.354	0.360	0.411	0.362	0.371
N	3615	3752	3556	3647	3475	3394	2612	5331

Table 8: Instrumental Variable Approach

This table report results of 2-SLS IV regressions. Column (1) reports results of the first-stage regressions. The dependent variable is *SC(Total)* estimated as the natural logarithm of 1 plus number of individuals with whom the CEO is connected via educational, employment, or other social links in BoardEx universe. Educational links are estimated as the number of individuals with whom the CEO attended the same university, graduated within 2 years of each other, and earned a similar type of degree. Employment links are constructed as the number of individuals with whom the CEO shares a common employment history. Other social links are defined as the number of individuals with whom the CEO is connected through social clubs or other non-profit organizations. *NSocialOrg* and $SC(Total)_{local}$ are the instrumental variables. *NSocialOrg* is the number of religious organizations, civic and social associations, business associations, political organizations, professional organizations, labor organizations, bowling centers, physical fitness facilities, public golf courses, and sport clubs in each county. $SC(Total)_{local}$ is the average number of social connections of other CEOs within 100 miles (excluding the CEO in question). Columns (2)-(8) report results of the second stage regressions. *SC(Total)* is replaced with the predicted values from the first-stage results. The dependent variables are *Volatility (Return), Volatility (ROA), Volatility (ROE), R&D/Sales, Diversification, Leverage, Working Capital*, respectively. *Volatility (Return)* is estimated as the volatility of monthly stock returns constructed over the window *t*+1 to *t*+3. *Volatility (ROA)* is estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+3. *Volatility (ROE)* is estimated as the volatility of a firm's quarterly return on asset, constructed over the window *t*+1 to *t*+3. Return on asset is the ratio of operating income to total assets and return on equity is the ratio of operating income to total stockholders' equity. *R&D/Sales* is the ratio of research and development expenses to total sales, estimated as the average over the window *t*+1 to *t*+3. *Diversification* is the measure of corporate diversification estimated following the methodology of Jacquemin and Berry (1979), constructed as the average over the window *t*+1 to *t*+3. *Leverage* is the ratio of total debt to total assets, measured as the average over the window *t*+1 to *t*+3. *Working Capital* is current assets minus current liabilities, scaled by total assets, measured as the average over the window *t*+1 to *t*+3. Control variables are defined in the Table 1 header. Robust standard errors, adjusted for heteroskedasticity and clustered at the firm level, are reported in brackets below the coefficients. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SC(Total)	Volatility (Return)	Volatility (ROA)	Volatility (ROE)	R&D/Sales	Diversifi-cation	Leverage	Working Capital
NSocialOrg	0.0508***							
	(0.00895)							
$SC(Total)_{local}$	0.0325**							
	(0.01458)							
$SC(Total)_{Predicted}$		0.0301***	0.0095***	0.1307***	0.5666***	-0.1620***	0.1900***	-0.0994***

		(0.00999)	(0.00284)	(0.04974)	(0.13531)	(0.05485)	(0.03736)	(0.02774)
Log (Age)	-0.0140	-0.0139***	-0.0015***	-0.0138***	-0.0777***	0.1065***	0.0124***	-0.0132***
	(0.00951)	(0.00063)	(0.00018)	(0.00334)	(0.00916)	(0.00341)	(0.00281)	(0.00208)
Log (Size)	0.3293***	-0.0225***	-0.0071***	-0.0567***	-0.2387***	0.0776***	-0.0392***	0.0008
	(0.00342)	(0.00330)	(0.00094)	(0.01644)	(0.04505)	(0.01810)	(0.01155)	(0.00857)
MB Ratio	0.0184***	-0.0001	0.0004***	0.0060***	0.0049	-0.0003	0.0004	0.0021***
	(0.00163)	(0.00022)	(0.00007)	(0.00168)	(0.00335)	(0.00111)	(0.00069)	(0.00044)
Sales Growth	-0.0181	0.0064***	0.0032***	0.0273***	0.0738**	0.0071*	0.0060	-0.0005
	(0.01426)	(0.00111)	(0.00043)	(0.00754)	(0.02917)	(0.00392)	(0.00548)	(0.00400)
Return	-0.0204**	-0.0001	-0.0011***	-0.0132***	-0.0186	0.0050*	0.0056	0.0051
	(0.00953)	(0.00068)	(0.00021)	(0.00372)	(0.01147)	(0.00297)	(0.00452)	(0.00327)
Debt/Equity Ratio	-0.0185***	0.0184***	-0.0007***	0.0664***	-0.0182***	-0.0127***		
	(0.00693)	(0.00059)	(0.00012)	(0.00402)	(0.00557)	(0.00241)		
Cash Surplus	-0.2697***	-0.1115***	-0.0200***	-0.2428***	-0.7130***	-0.2056***	-0.2684***	0.1397***
	(0.05289	(0.00467	(0.00167	(0.03019	(0.08473	(0.02022	(0.02824	(0.02075

	)	)	)	)	)	)	)	)
Log (CEO Age)	-0.3394***	-0.0122***	0.0040***	0.0449**	0.1701***	0.0174	0.0509***	-0.0232*
	(0.04257)	(0.00434)	(0.00125)	(0.02140)	(0.06274)	(0.02295)	(0.01602)	(0.01211)
Log (CEO Tenure)	-0.0886***	-0.0004	-0.0002	-0.0021	0.0479***	-0.0080	0.0052	-0.0010
	(0.00664)	(0.00099)	(0.00029)	(0.00508)	(0.01351)	(0.00541)	(0.00395)	(0.00293)
CEOGD	-0.2887***	0.0050	0.0004	0.0322	0.2470***	-0.0398**	0.0838***	-0.0662***
	(0.03585)	(0.00376)	(0.00113)	(0.02086)	(0.05006)	(0.02012)	(0.01439)	(0.01056)
Constant	3.7239***	0.2232***	0.0106	-0.4368**	-1.7373***	0.5672**	-0.5652***	0.9686***
	(0.22722)	(0.04404)	(0.01246)	(0.21452)	(0.59998)	(0.24512)	(0.16322)	(0.12159)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-sq	0.39	0.337	0.201	0.012	0.05	0.24	0.78	0.298
N	29356	29356	29356	29356	29356	29356	29356	29356

Table 9: CEO Social Capital, Corporate Risk-taking and Risk-adjusted Value

This table reports estimates of coefficients from an OLS regression. The dependent variable is the excess one year holding period return, estimated as the difference between the actual return over year *t* to *t*+1 and expected return estimated using Fama-French four factor model. *Abnormal Volatility* is predicted abnormal volatility attributable to CEO social capital (constructed using *Volatility(Return)* - volatility of monthly stock returns over the window *t*-1 to *t*-3). Control variables are defined in the Table 1 header. In Columns (1)-(2), (3)-(4), (5)-(6), (7)-(8), and (9)-(10) the sample is partitioned based on median values of *CEO Pay Slice*, *Free Cash Flow*, *External Financing*, *Information Asymmetry* and *Investment Efficiency*, respectively. *CEO Pay Slice* is the fraction of the aggregate compensation of the top-five executive team captured by the CEO. *Free Cash Flow* is estimated following Lehn and Poulsen (1989). *External Financing* is the annual change in book value of equity plus the change in deferred taxes minus the change in retained earnings, all scaled by lagged asset. *Information Asymmetry* is the absolute value of analysts' earnings forecast error for the fiscal year, estimated as the absolute value of actual earnings minus the earnings forecast scaled by the stock price. *Investment Efficiency* is estimated as the residual from a simple investment model (Biddle et al., 2009) that predicts the level of investment based on growth opportunities (measured by sales growth). Positive residual, as reflected in the error terms of the investment model, represent overinvestment and negative residual – underinvestment. Robust standard errors, adjusted for heteroskedasticity and clustered at the firm level, are reported in brackets below the coefficients. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	CEO Pay Slice		Free Cash Flow		External Financing		Information Asymmetry		Investment Efficient	
	High	Low	High	Low	High	Low	High	Low	Overinvest	Underinvest
Abnormal Volatility	0.0028	0.0442**	0.0106	0.0169	0.0269**	0.0095	0.0485***	0.0037	0.0121	0.0232*
	(0.01380)	(0.02098)	(0.01267)	(0.01143)	(0.01273)	(0.01353)	(0.01516)	(0.01561)	(0.01237)	(0.01235)
Log (Age)	-0.0065	-0.0045	-0.0203***	0.0020	-0.0245**	-0.0155	-0.0121	-0.0159*	-0.0140*	-0.0229**
	(0.00973)	(0.01389)	(0.00732)	(0.01143)	(0.01081)	(0.00955)	(0.01348)	(0.00924)	(0.00815)	(0.01122)
Log (Size)	-0.0027	-0.0223	-0.0107	-0.0128	-0.0113	-0.0164	-0.0200	-0.0112	-0.0150	-0.0107*

		***	***	**	***	***	***	***	***	**
	(0.00521)	(0.00496)	(0.00363)	(0.00533)	(0.00325)	(0.00310)	(0.00456)	(0.00338)	(0.00331)	(0.00290)
MB Ratio	0.0015	0.0003	0.0013	-0.0058***	-0.0012	-0.0031	0.0012	0.0005	-0.0027	-0.0019
	(0.00301)	(0.00214)	(0.00149)	(0.00214)	(0.00245)	(0.00209)	(0.00414)	(0.00165)	(0.00194)	(0.00233)
Sales Growth	-0.0228	0.0370	0.0097	0.0279	0.0221	0.0235	0.0520*	0.0016	0.0175	0.0132
	(0.02885)	(0.03761)	(0.02566)	(0.01937)	(0.01635)	(0.04390)	(0.03108)	(0.02207)	(0.01970)	(0.02313)
Debt/Equity Ratio	0.0249**	0.0298***	0.0322***	0.0306***	0.0275***	0.0360***	0.0565***	0.0615***	0.0440***	0.0269***
	(0.00994)	(0.01041)	(0.00801)	(0.01024)	(0.00942)	(0.00856)	(0.01011)	(0.02237)	(0.01032)	(0.00837)
Cash Surplus	-0.0378	-0.1262	-0.0005	-0.0185	0.0596	-0.0075	-0.0381	-0.0586	-0.1066	0.0871
	(0.08783)	(0.18433)	(0.08149)	(0.06009)	(0.05769)	(0.09015)	(0.10208)	(0.07855)	(0.09516)	(0.05986)
Log (CEO Age)	-0.1742***	0.0419	-0.0739**	-0.0246	-0.0658	-0.1087**	-0.0810	-0.0940**	0.0486	-0.1575***
	(0.05289)	(0.06836)	(0.03644)	(0.04690)	(0.04377)	(0.04414)	(0.05791)	(0.03919)	(0.04232)	(0.04105)
Log (CEO Tenure)	0.0089	0.0091	0.0017	0.0007	-0.0005	0.0087	0.0016	0.0086	0.0036	0.0088
	(0.00767)	(0.01057)	(0.00528)	(0.00718)	(0.00681)	(0.00679)	(0.00865)	(0.00602)	(0.00625)	(0.00677)
CEOGD	-0.0090	0.0921**	0.0420	0.0420	0.0482	0.0489	0.0786**	0.0154	0.0320	0.0416
	(0.04376)	(0.04098)	(0.03305)	(0.03567)	(0.03304)	(0.03225)	(0.03125)	(0.04448)	(0.03277)	(0.03246)
Constant	0.9587***	0.1296	0.6030***	0.5767***	0.6930***	0.7956***	0.7652**	0.7649***	0.1369	1.2144***

	(0.22063)	(0.37989)	(0.17220)	(0.20296)	(0.18764)	(0.19105)	(0.30442)	(0.16213)	(0.17900)	(0.22791)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R-sq	0.114	0.082	0.116	0.111	0.105	0.103	0.101	0.108	0.104	0.102
N	5996	6038	13320	13097	13290	13524	10066	10205	13864	14862

Highlights

Ms Ref No: CORFIN D-16-00324
"CEO Social Capital, Risk-Taking, and Corporate Policies"

- Positive association between CEO social capital and aggregate corporate risk-taking
- CEO social capital increases the riskiness of specific corporate investment and financial policies
- Social capital induces CEOs to take more risk which is value-enhancing conditional on a set of moderating variables
- The effect of social capital on risk-taking is stronger for firms with low managerial entrenchment, low free cash flows, high external equity financing, high information asymmetry and previous underinvestment.